# ECI v7.6 amendment: Phase 3.E synthesis: ECI production posterior, $\xi_\chi$ rail diagnostic, $N = p^2$ interlinking, $B_{\text{Higgs}}$ modular-geometry audit, Wolf–ECI–AKW three-way contest, SUSY NO-GO, and honest tensions

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**Abstract**

The v7.4 amendment of the Extended Cosmological Index (ECI) research programme [**?**] relaxed the strict $S$-fixed point axiom $H_5 \to H_5'$ (CM-anchored attractor) and replaced the Damerell–Coates–Sinnott bridge $H_7 \to H_7'$ by a two-part bridge (integer-point Damerell ladder + BDP $p$-adic complement). The v7.5 amendment (2026-05-05) recorded six concrete extensions, one retraction, and one structural reframing. The **present v7.6 amendment** (2026-05-06, Phase 3.E) adds eight further deliverables from sub-agents M20–M24 and O1: (i) the first ECI production posterior on real DESI DR1/Pantheon+/Planck 2018 data (§**??**), with 4-chain NUTS on an RTX 5060 Ti, $\hat{R} = 1.0001$, ESS = 30,683, and $H_0 = 68.51 \pm 0.14$ (Planck-consistent); (ii) the $\xi_\chi$ rail diagnostic ($\xi_\chi = -0.0240 \pm 1.8 \times 10^{-5}$) confirmed as a real data signal rather than a prior artefact (M24; §**??**); (iii) the $N = p^2$ interlinking meta-finding that unifies M13's three arithmetic findings for the CM newform 4.5.b.a via the Steinberg-edge property (M22; §**??**); (iv) the $B_{\text{Higgs}} = 1.077$ modular-geometry audit from a clean amplitude computation (M23; §**??**); (v) the Wolf–ECI–AKW three-way contest framework (M9 reframe + arXiv:2604.08449 Antusch–King–Wang 2026; §**??**); (vi) a Conjectural Adelic Katz paper-2 framework (M3+M13+M20+M22 synthesis; §**??**); (vii) the SUSY MSSM NO-GO decision with split-SUSY conditional (O1; §**??**); and (viii) an honest tensions audit (Wolf25 $\xi$ tension, $\Sigma m_\nu$, 144.5.g.a alternative CM; §**??**). Hallu count: 85 in / 85 out.

**Retraction (A46):** the v7.4 "LYD20 unified scaffold with $\tau$ pinned at $\tau_S = i$" graft is empirically REFUTED. A 19-observable joint quark+lepton refit (PDG 2024) yields $\chi^2$ penalty $\approx 99{,}061\times$ versus the LYD20 published best fit at $\tau_{\text{LYD20}} = -0.21 + 1.52\,i$ – five orders of magnitude beyond the $\sim 1.8\times$ tolerance estimated by the earlier A42 scoping. v7.5 therefore adopts Option C: NPP20 lepton-only at $\tau_S = i$ (4–6 free parameters for 9 observables) plus the G1.12.B SU(5) $5_H + 45_H$ modular $f^{ij}(\tau = i)$ campaign for the quark sector, DROPPING the unified scaffold.

**Grafts:** (1) a stand-alone Cassini-wall section grafting the Karam–Sánchez López–Terente Díaz analytical PPN reference (KSTD 2026, `arXiv:2604.16226`), with v7.5-P Palatini sub-branch registered as a conditional 2027-decision direction; (2) the King–Wang multi-modulus dS-trap (KW 2023, `arXiv:2310.10369`) supplying a string-cosmological mechanism for $\tau \to i, \omega$ vacua (closes A14's "WHY $\tau = i$?" gap); (3) the dMVP26 canonical-Kähler quark hierarchy at $\tau = i$ (A48 PARTIAL graft) with $J^q_{CP} = 0$ cost flagged; (4) leptogenesis from the CSD$(1 + \sqrt{6})$ Littlest Modular Seesaw at $\tau_S = i$ (A55) yielding the structural signature $(n-1)^2 = 6$ in $\epsilon_1$ scaling and Planck-compatible $Y_B^{\text{calib}} = 0.872 \times 10^{-10}$.

**Structural reframings:** (5) the A42 "two-$\tau$ is 60× worse" claim is corrected per A49 autopsy of Du–Wang `arXiv:2209.08796` – the comparison was a category error; two-$\tau$ is GENERICALLY HELPFUL empirically. v7.5 keeps single-$\tau$ on STRUCTURAL grounds (Du–Wang's GUT matter-unification orbifold-BC reduction), not numerical $\chi^2$. (6) The A54 BC×CM at $\beta = 2\pi$ chain (Connes–Marcolli–Ramachandran 2005, `arXiv:math/0501424`) is honestly downgraded: it does NOT bypass $H_6$ ($\chi_4$ nebentypus restriction) and joins the Galois-descent chain (P-NT) as a SECOND $H_6$-co-dependent partial derivation, not two independent analytical derivations.

We collect seven experimental disambiguators 2027–2045 (NA62, JUNO, DUNE, KamLAND-Zen, CMB-S4, Belle II, Hyper-K) in Section **??** and list open questions in Section **??**. The arithmetic-modular core ($f_{4.5.b.a}$, $\mathbb{Q}(i)$, Damerell ladder, CSD$(1+\sqrt{6})$, G1.12.B M1–M6) is preserved across the v7.4 → v7.5 transition.

**Honest negative result (A72, post-submission):** A null test (A72, 2026-05-06) finds that the $K = \mathbb{Q}(i)$ Damerell-ladder numerology is statistically indistinguishable from random rational ladders on the full PMNS+lepton+quark observable space ($Q(i) = 358$ vs null $= 366 \pm 25$, $\sigma = -0.33$, $P = 0.63$, strict cut). The Damerell ladder survives as an *arithmetic anchor* for $f_{4.5.b.a}$ (LMFDB 4.5.b.a, Hecke $H_1$, $\chi_4$ nebentypus) but does NOT constitute a predictive structural input for low-energy fermion observables beyond what random rationals achieve. Sections **??** and **??** document this downgrade.

# 1 Introduction: from v7.4 to v7.6

The v7.4 amendment [**?**] closed three open questions of the v7.3 axiomatic restart of ECI [**?**]: it relaxed the strict $S$-fixed point $H_5$ to a CM-anchored attractor $H_5'$ ($|\tau - i| \lesssim 0.2$, supported by a $30 \times 30$ $\chi^2$ scan W1 yielding $\tau^* = -0.1897 + 1.0034\,i$ at $\chi^2/\mathrm{dof} = 1.05$); replaced the half-integer Damerell–Coates–Sinnott bridge $H_7$ – refuted by the G1.15 audit on parity grounds for an odd-weight CM newform – by an integer-point Damerell ladder $H_7'$ ($\alpha_m \in \{1/10, 1/12, 1/24, 1/60\}$ for $m \in \{1,2,3,4\}$) with parameter-free Cardy hits $\rho_{\mathrm{Cardy}}(c = 1) = 1/12$ (free boson) and $\rho_{\mathrm{Cardy}}(c = 1/2) = 1/24$ (Ising); and recorded six wave 4–5 follow-ups (A14, A16, A17, A22, A24, A26).

The present v7.5 amendment records the post wave 4–5 development of 2026-05-05 (evening and night), comprising one retraction (A46), four grafts (A47, A48, A50, A55), and two structural reframings (A49, A54). Section **??** grafts the Karam–Sánchez López–Terente Díaz analytical PPN reference for the Cassini wall (KSTD 2026, `arXiv:2604.16226`) and registers a Palatini sub-branch v7.5-P as a conditional 2027-decision direction. Section **??** collects the v7.5 modular flavour body in Option-C form (NPP20 lepton + G1.12.B quark), with subsections covering: G1.12.B SU(5) M1–M6 closure including the M5 partial-width computation (Sec. **??**); the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw at $\tau_S = i$ (Sec. **??**); the A24 $\sqrt{6}$ Galois-rationality closure (Sec. **??**); the A47 King–Wang dS-trap physical mechanism for $\tau = i, \omega$ (Sec. **??**); and the A55 leptogenesis $(n-1)^2 = 6$ structural signature (Sec. **??**). Section **??** records the A48 partial graft of the dMVP26 canonical-Kähler quark hierarchy at $\tau = i$ with the $J^q_{CP} = 0$ cost. Section **??** re-frames the two-$\tau$ debate per A49. Section **??** preserves the v7.4 sub-algebra $H_1$ closure, Damerell ladder, and A17 numerological CKM hits *with A72 downgrade*. Section **??** records the conditional analytical $\tau$ derivations (A54 BC×CM and Galois descent in P-NT) honestly framed as $H_6$-co-dependent. Section **??** lists seven experimental disambiguators 2027–2045. Section **??** records the A72 honest negative result. Section **??** collects open questions and the v7.5-P Palatini sub-branch decision matrix.

**A46 retraction (the leading v7.4 → v7.5 change).** The v7.4 strategic synthesis (commit `6.0.53.5`) had assumed A42's scoping estimate that the LYD20 unified quark+lepton scaffold of [**?**], when $\tau$ is strictly pinned at the $S$-fixed-point $\tau_S = i$, would carry an empirical $\chi^2$ penalty of order 1.8× versus LYD20's published best fit at $\tau_{\mathrm{LYD20}} = -0.2123 + 1.5201\,i$. A46 (this volume) ran the explicit 19-observable joint Q+L refit (10 free LYD20 parameters after pinning $\tau$):

| Fit configuration | $\tau$ | $\chi^2_{\min}$ (19 obs) | Penalty |
|---|---|---|---|
| LYD20 published best | $-0.2123 + 1.5201\,i$ | **1,009** | reference (0×) |
| W1 attractor (v7.4) | $-0.1897 + 1.0034\,i$ | 878,990 | +870× |
| **Strict $\tau_S = i$ (v7.4 pinning hypothesis)** | $0 + 1.0\,i$ | **$\sim 10^8$** | **+99,061×** |

Table 1: A46 single-$\tau$ refit of the LYD20 unified Q+L scaffold, PDG 2024 covariances, 19 observables (8 fermion masses + 4 CKM moduli + 4 PMNS angles + $\delta_{CP}$ + 2 mass-squared splittings). The v7.4 "pin $\tau$ at $i$" hypothesis is REFUTED at $\sim 5$ orders of magnitude beyond the A42-scoped tolerance. Full results JSON in `notes/eci_v7_aspiration/A46_LYD20_GRAFT/lyd20_pinned_results.json`.

The catastrophic penalty traces to the V2 no-go mechanism [**?**]: at $\tau_S = i$ the $\{d^c, s^c\}$-block of the down-quark Yukawa becomes near-rank-1 by an $S$-collinearity lemma, killing $\sin\theta_C$ and propagating into PMNS through the LYD20 unified-scaffold's structural quark–lepton mixing.

**Honest revision (Option C of A46).** v7.5 abandons the unified-scaffold pinning and adopts:

- *Lepton sector:* NPP20 model [**?**] ($S'_4 \hookrightarrow \widetilde{\Gamma}(4)$ doublet/triplet assignments, 4–6 free parameters for 9 observables) at the strict $S$-fixed point $\tau_S = i$, in CSD$(1+\sqrt{6})$ Littlest Modular Seesaw alignment.
- *Quark sector:* G1.12.B SU(5) $5_H + 45_H$ modular construction with $f^{ij}(\tau = i)$ built from row-by-row LYD20-Model-VI contractions (rows $u, c, t$ at weights $1, 2, 5$, irreps $\hat{3}'$, $\mathbf{3}$, $\hat{3}$); modular invariance inherited; M1–M2 closed (this volume) and M3–M6 forecast at 3.5–5 sub-agent months. Quark hierarchies via dMVP26 canonical-Kähler at $\tau = i$ (Section **??**).

The arithmetic anchor $f_{4.5.b.a}$ (LMFDB CM newform 4.5.b.a, weight 5, CM by $\mathbb{Q}(i)$) and the integer-point Damerell ladder $H'_7$ are unchanged across the A46 retraction. The W1 attractor result (v7.4) is now interpreted as a partial-fit structure on the 7 fermion mass ratios alone, NOT a unified Q+L claim.

**A49 correction propagated.** The A42 row-5 claim "two-$\tau$ is 60× worse" was a category error: it compared the single-$\tau$ Sample IX′ sweet spot ($\chi^2_{\text{tot}} = 1.58$) of Du–Wang Scenario III against the dual-$\tau$ generic-bad worst sub-scenarios ($\chi^2 \approx 95$). Du–Wang [**?**, Tab. 5, §4] themselves report two-$\tau$ *improves* the lepton fit 4–16× in every sub-scenario where they tabulate both. Section **??** reframes the v7.5 single-$\tau$ choice as a STRUCTURAL preference (GUT matter unification, Du–Wang's $M_4 \times S^1/Z_2$ orbifold-BC reduction) rather than an empirical $\chi^2$ argument.

**Phase 3.E deliverables (v7.5 → v7.6, 2026-05-06).** The v7.6 amendment adds eight new sections driven by sub-agents M20–M24 and O1. In brief: 14 sub-agent audits have now been completed (M1–M24 plus O1); the Lakatos hard core has shrunk to 5 falsifiable claims; Conjecture M13.1 has reached ANT (*Adelic Number Theory*) tier and motivates a standalone math.NT paper; the ECI production posterior has CONVERGED on real DESI DR1 + Pantheon+ + Planck 2018 data with $\hat{R} = 1.0001$ (4 chains, 8000 samples each); and 5 LMP/BLMS papers are SUBMISSION-READY with the 6th (proton-decay PRD) awaiting M5 redo with full modular $f^{ij}$. A sixth new constraint is the $\xi_\chi = -0.0240 \pm 1.8 \times 10^{-5}$ rail finding, now confirmed as a real data signal consistent with RG running from $\xi(M_Z) \simeq +0.001$ down to $\xi(M_{\text{GUT}}) \simeq -0.029$ (sub-agent A73; §**??**).

**Hallucination policy.** This amendment reuses the v74_amendment_v2 verified bibliography verbatim (28 entries, all live-verified 2026-05-05 via the arXiv API) and adds 8 new live-verified

entries for the v7.5 grafts (KSTD26, KW23, KMSR18, CMR05, Bertotti03, KZ24, CMBS4, JUNO22). The v7.6 amendment adds further bibentries only if live-verified; all new citations are flagged at their first appearance. The hallucination counter is held at 85. Author-name errata respected throughout. Mistral large-latest STRICT-BANNED. Computational substrate: PC RTX 5060 Ti (cosmopower-jax 386 predictions/s, JAX named_shape patch); LMFDB queries cached 2026-05-05–06.

## 2 Cassini wall and Palatini formulation

The NMC branch of ECI v7.4 enforces a Cassini compliance wall $|\xi_\chi|(\chi_0/M_{\rm P})^2 \lesssim 6\times10^{-6}$, recovered from the standard Solar-System bound $|\gamma-1|\le 2.3\times10^{-5}$ (Cassini Doppler tracking, [**?**]) translated into the Jordan-frame Lagrangian $\mathcal{L}\supset -\frac{1}{2}\xi_\chi R\chi^2$ (Faraoni convention; cf. [**?**], App. A). In this section we graft an explicit analytical PPN reference for the wall and register a Palatini sub-branch (ECI v7.5-P) as a conditional 2027-decision research direction.

**Analytical Cassini-wall reference (KSTD 2026).** Karam, Sánchez López, and Terente Díaz [**?**] (`arXiv:2604.16226`) recently derived a unified post-Newtonian treatment of scalar–tensor gravity with arbitrary non-minimal coupling $A(\Phi)$, kinetic function $B(\Phi)$, and potential $V(\Phi)$, in BOTH the metric and Palatini formalisms. In their notation, with $\delta_{\rm P}=0$ (metric) or $\delta_{\rm P}=1$ (Palatini) and $A_n, B_n, V_n$ the Taylor coefficients of $A$, $B$, $V$ about the cosmological background, the effective scalar mass and PPN parameter $\gamma$ at distance $r$ from a point source read

$$m_\varphi^2 = \frac{16\pi G\,A_0\,V_2}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_{\rm P})}, \qquad \text{(KSTD 3.38)}$$

$$G_{\rm eff}(r) = \frac{G}{A_0}\left[1+\frac{A_1^2\,e^{-m_\varphi r}}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_{\rm P})}\right], \qquad \text{(KSTD 3.41)}$$

$$\gamma(r) = 1 - \frac{2A_1^2\,e^{-m_\varphi r}}{16\pi G\,A_0\,B_0+3A_1^2(1-\delta_{\rm P})+A_1^2\,e^{-m_\varphi r}}. \qquad \text{(KSTD 3.42)}$$

Specialising to a non-minimally coupled scalar with canonical kinetic term ($B_0=1$, $B_1=0$) yields the metric Yukawa coupling $\alpha = A_1^2/(16\pi G A_0+3A_1^2)$ and the Palatini one $\hat\alpha = A_1^2/(16\pi G A_0)$ (KSTD eqs. 4.31–4.32); the analogous expression for $\beta(r)$ is given by KSTD eq. (3.67).

**Consistency check with the ECI v7.4 Cassini wall.** For the ECI Lagrangian $-\frac{1}{2}\xi_\chi R\chi^2$ expanded about $\chi_0$, one identifies $A_0=1-\xi_\chi\chi_0^2/M_{\rm P}^2$ and $A_1=-\xi_\chi\chi_0/M_{\rm P}$, so that $A_1^2/(16\pi G A_0)\simeq 4\,\xi_\chi^2(\chi_0/M_{\rm P})^2$ to leading order. In the metric formalism with strong Yukawa suppression at Solar-System scale, KSTD eq. (3.42) reduces to $|\gamma-1|\approx 2A_1^2/(16\pi G A_0)\simeq 8\xi_\chi^2(\chi_0/M_{\rm P})^2$, and the Cassini bound $|\gamma-1|\le 2.3\times10^{-5}$ translates into $\xi_\chi^2(\chi_0/M_{\rm P})^2\lesssim 3\times10^{-6}$, i.e. $|\xi_\chi|(\chi_0/M_{\rm P})^2\lesssim 6\times10^{-6}$ at the small-$\xi_\chi$ root — in agreement with the wall enforced by the v7.4 NMC posterior [**?**, App. A]. The ECI v7.4 mean $\xi_\chi=-3\times10^{-5}$ (Levier 1B) sits $\sim 10^2$ orders of magnitude inside the wall, so the NULL at Cassini is structurally enforced rather than fine-tuned.

**Forward-pointing remark: the v7.5-P Palatini sub-branch.** KSTD show explicitly that, at fixed $(A_0, A_1, B_0)$, the Palatini formalism admits a substantially wider allowed region in $V_2$ than the metric one — by up to $\sim 10$ orders of magnitude in the regime of small $A_0$ and large $|A_1|$ (KSTD Fig. 2 and §4.2.1). The mechanism is geometric: in Palatini the screening length is $\hat m_\varphi^{-1} = V_2^{-1/2}$, independent of $A_0\,B_0$, leading to stronger Yukawa suppression of the fifth-force term in $\gamma(r)$. As an open research direction, an *ECI v7.5-P Palatini sub-branch* (sketch

in Section **??**) may relax the Cassini wall by several orders of magnitude while preserving the cosmological NMC signal at $\xi_\chi \sim O(10^{-2})$ – re-opening contact with detection-side analyses such as [**?**] that lie outside the current ECI metric-formalism wedge. The arithmetic-modular anchors of v7.4–7.5 ($f_{4.5.\mathrm{b.a}}$, $\tau$-attractor, Damerell ladder, CSD($1+\sqrt{6}$), G1.12.B) are formalism-independent and carry over unchanged. *We do not commit to v7.5-P in the present amendment*; the metric formalism remains the conservative baseline, and the Cassini NULL falsifier listed in Section **??** applies to v7.5-metric only.

# 3 Axiomatic foundation of ECI v7.5

This section gives formal statements of the 11 axioms of ECI v7.5, with Lakatos-honest status tags (Established / Working-Confirmed / Postulate / Retracted), source references, and explicit falsifiers. The structure is: $H_1$–$H_4$ (operator-algebra hard core), $H'_5$–$H'_7$ (modular-flavour protective belt, amended in v7.4), and $H'_8$–$H'_{11}$ (new v7.5 grafts: A47/A48/A50/A55 and the A46 retraction of the LYD20 unified scaffold).

## 3.1 Hard core (operator-algebra framework)

**Axiom** ($H_1$ (refined post-A79), type-II$_\infty$ crossed product — partitioned)**.** *The crossed-product von Neumann algebra of observables for a gravitational subregion with a suitable dynamical clock, obtained from the type-III$_1$ algebra $\mathcal{M}(\mathcal{D})$ by gauging the modular automorphism group of a faithful normal state, is of type II$_\infty$: it admits no minimal projection and carries a faithful semifinite normal trace. A79's post-CLPW literature survey (22 papers, arXiv-API live-verified 2026-05-05) finds this algebra* established *in exactly 4 sub-classes and* conjectural *in at least 4 further sub-classes; the 4 conjectural sub-classes lie entirely outside the ECI Cassini-clean parameter regime.*

**4 ESTABLISHED sub-classes (ECI uses $H_{1,\mathrm{a}}$ and $H_{1,\mathrm{d}}$):**

$H_{1,\mathrm{a}}$ **Static-patch de Sitter.** (Established.) Chandrasekaran–Longo–Penington–Witten 2023 [**?**] (arXiv:2206.10780). Full proof: York-time comoving clock, complete type II$_\infty$ classification.

$H_{1,\mathrm{b}}$ **Quasi-de Sitter / slow-roll inflation.** (Established.) Chen–Penington 2024 (arXiv:2406.02116); Speranza 2025 (arXiv:2504.07630). Type II$_\infty$ with rolling inflaton clock; compact no-Killing-symmetry wedges also classified.

$H_{1,\mathrm{c}}$ **FLRW asymptotically-dS in the inflationary past.** (Established.) Kudler-Flam–Leutheusser–Satishchandran 2024 [**?**] (arXiv:2406.01669, "Algebraic Observational Cosmology"): first non-static-patch construction; gravitationally-dressed comoving observer algebra for FLRW with inflaton zero-mode clock; all states mixed, vN entropy well-defined.

$H_{1,\mathrm{d}}$ **4D flat FRW radiation/matter-dominated era, $\xi = 1/6$ conformally-coupled scalar.** (Established.) ECI-internal [**?**] (`paper/frw_typeII_note/frw_note.tex`, Remondière 2026-05-02): conformal pullback at $\xi = 1/6$ gives type II$_\infty$ for the comoving past-light-cone diamond in both the radiation-dominated and matter-dominated eras. *Primary ECI operative regime.* Cassini-clean: $\xi = 1/6$ is the conformal value; IR pathology of the $\xi = 0$ massless scalar on dS (Kudler-Flam–Prabhu–Satishchandran 2025, arXiv:2503.19957) directly motivates this choice.

**4 CONJECTURAL sub-classes (outside ECI regime; do NOT weaken the hard core):**

$H_{1,\mathrm{e}}$ **Generic flat ΛCDM with Λ.** (Conjectural.) No comoving Killing vector; no asymptotic-dS clock for the radiation/matter eras; KFLS24 transport to the matter epoch blocked; single-observer algebra bridging the matter-to-Λ crossover unproven.

$H_{1,\mathrm{f}}$ **Bianchi backgrounds.** (Conjectural.) Anisotropy destroys conformal-pullback cleanness; no gravitationally-dressed observer construction exists.

$H_{1,\mathrm{g}}$ **Massive / interacting fields beyond $O(m^2)$.** (Conjectural.) Mass term breaks $\xi = 1/6$ conformal coupling; no type $\mathrm{II}_\infty$ classification for interacting matter on FRW is known.

$H_{1,\mathrm{h}}$ **Single-observer description bridging the matter-to-Λ crossover.** (Conjectural.) Algebra changes character at the transition; no unified crossed-product construction spanning both epochs.

**Status.** Established in $H_{1,\mathrm{a}}$–$H_{1,\mathrm{d}}$. Conjectural in $H_{1,\mathrm{e}}$–$H_{1,\mathrm{h}}$ (outside ECI regime; do NOT weaken ECI). **References:** CLPW 2023 [**?**] (arXiv:2206.10780); KFLS 2024 [**?**] (arXiv:2406.01669, live-verified 2026-05-05 via arXiv API, title: "Algebraic Observational Cosmology", authors: Kudler-Flam, Leutheusser, Satishchandran); ECI-internal [**?**]; A79 survey SUMMARY.md (22 papers, 0 fabrications, hallu count held at 85). **Falsifier.** Constructive identification of a minimal projection in $\widetilde{\mathcal{M}}(\mathcal{D})$ for the static-patch dS algebra at the BW temperature (kills $H_{1,\mathrm{a}}$); or proof that conformal pullback at $\xi = 1/6$ in the radiation/matter era fails to produce a semifinite trace (kills $H_{1,\mathrm{d}}$).

**Axiom** ($H_2$, KMS at $\beta = 2\pi$ (Bisognano–Wichmann))**.** *The state $\omega$ is KMS at $\beta = 2\pi$ for the modular flow $\sigma_t^\omega$; $K_R$ generates a flow of period $2\pi$ and $\omega(A\,\sigma_{i\beta}^\omega(B)) = \omega(BA)$ holds for all $A, B \in \widetilde{\mathcal{M}}(\mathcal{D})$.*

**Status.** Established (Tomita–Takesaki theory; $\beta = 2\pi$ selects the Hartle–Hawking / Bunch–Davies state).

**Axiom** ($H_3$, microlocal functoriality ($\mathrm{PH}_k$ axiom A6))**.** *All constructions are locally-covariant-functorial (Brunetti–Fredenhagen–Verch 2003); $\mathrm{PH}_k$ satisfies Kashiwara–Schapira axiom A6.*

**Status.** Established (P-KS standalone paper).

**Axiom** ($H_4$, structural inheritance (vacuous exclusion clause))**.** *No structure beyond $H_1$–$H_3$: no preferred vacuum, no Connes spectral triple, no boundary CFT, no global extremisation. Every dimensionful observable is determined only modulo $\mathcal{S} = \mathbb{R}_{>0} \rtimes \mathrm{Out}(\widetilde{\mathcal{M}})$ (Theorem 1 of [**?**]).*

**Status.** Established (vacuous; defines Theorem 1 scope).

### 3.2 Modular-flavour protective belt (amended in v7.4)

**Axiom** ($H_5'$, CM-anchored attractor)**.** $|\tau - i| \lesssim 0.2$*; W1 scan: $\tau^* = -0.1897 + 1.0034\,i$, $\chi^2/\mathrm{dof} = 1.05$ on seven fermion observables.*

**Status.** Working-Confirmed (W1 scan 2026-05-05).

**Axiom** ($H_6$, Hecke restriction $\mathcal{H}_1$)**.** *Mass-matrix elements close under $\mathcal{H}_1 = \{T(p) : p \equiv 1 \bmod 4\}$; obstruction at $p \equiv 3 \bmod 4$ from $\chi_4$ nebentypus of LMFDB `4.5.b.a`, verified $p \leq 113$.*

**Status.** Established (5-prime over-determination).

**Axiom** ($H_7'$, integer-point Damerell ladder (notation convention post-A72)). *For $f_{4.5.b.a} \equiv$ `4.5.b.a` (CM by $\mathbb{Q}(i)$, weight 5, level 4) and $\Omega_K$: $\left\{L(f_{4.5.b.a}, m)\pi^{4-m}/\Omega_K^4\right\}_{m=1}^4 = \{1/10, 1/12, 1/24, 1/60\}$; Cardy densities $\rho_{\text{Cardy}}(c = 1) = 1/12$, $\rho_{\text{Cardy}}(c = 1/2) = 1/24$ parameter-free.* ***Post-A72 reclassification:*** *$H_7'$ is a* notation convention *for the Damerell-derived rationals characterising CM L-values at $\mathbb{Q}(i)$ (mathematically rigorous via Hurwitz/Chowla–Selberg/Damerell/Shimura anchoring),* ***not*** *a falsifiable structural input for low-energy fermion observables (cf. Section **??**, Section **??**).*

**Status.** Notation convention post-A72 null test (downgrade from Established in v7.4); the *mathematical* ladder is rigorous (mpmath dps=60; [**?**, **?**, **?**, **?**, **?**]); the *predictive* reading on low-energy fermion observables is retired (A72: $\sigma = -0.33$, $P_{\text{null}} = 0.63$).

### 3.3 New v7.5 grafts

**Axiom** ($H_8'$, Karam–Palatini Cassini wall (new v7.5)). *Cassini wall $|\xi_\chi|(\chi_0/M_{\text{P}})^2 \lesssim 6 \times 10^{-6}$ recovered from KSTD 2026 [**?**] (arXiv:2604.16226) eq. (3.42) in the metric formulation; Palatini defines v7.5-P (not committed).*

**Status.** Postulate (A50 graft).

**Axiom** ($H_9'$, King–Wang dS-trap mechanism (new v7.5)). *KW 2023 [**?**] (arXiv:2310.10369) kinematically selects $\tau \in \{i, \omega\}$; A14 cross-K test selects $\tau = i$ uniquely.*

**Status.** Postulate (A47 graft).

**Axiom** ($H_{10}'$, leptogenesis $(n-1)^2 = 6$ fingerprint (new v7.5)). *At $\tau_S = i$ in CSD$(1+\sqrt{6})$: $(n-1)^2 = 6$ and $Y_B^{\text{CSD}(1+\sqrt{6})}/Y_B^{\text{CSD}(3)} = 3/2$.*

**Status.** Working-Confirmed (A55; [**?**] eqs. 22–25 live-verified; 7/512 grid points inside Planck $\pm 1\sigma$).

**Axiom** ($H_{11}'$, Option C scaffold; LYD20-unified retracted (new v7.5)). *LYD20 unified at $\tau = i$:* ***Retracted*** *(A46; 99,000× $\chi^2$ penalty). Option C: NPP20 [**?**] lepton (4–6 parameters, 9 observables) at $\tau_S = i$; G1.12.B SU(5) $f^{ij}(\tau = i)$ for quarks [**?**] (M1+M2 PASS; M3–M6 forecast).*

**Status.** Working-Confirmed post-retraction (A22 M1 PASS 4/4; A26 M2 PASS 5/5).

### 3.4 Lakatos status table (post-A79 H1 partition)

## 4 Modular flavour body (Option C: NPP20 lepton + G1.12.B quark at $\tau = i$)

Per the A46 retraction (Section **??**), the v7.5 modular flavour body splits into: a lepton sector following NPP20 [**?**] at the strict $S$-fixed point $\tau_S = i$, in CSD$(1+\sqrt{6})$ Littlest Modular Seesaw alignment (Section **??**); and a quark sector following the G1.12.B SU(5) $5_H + 45_H$ modular construction (Section **??**) with hierarchies supplied by dMVP26 canonical-Kähler at $\tau = i$ (Section **??**). The A47 King–Wang multi-modulus dS-trap supplies the string-cosmological mechanism that selects $\tau \in \{i, \omega\}$ at the level of the modulus potential (Section **??**); the A14 cross-$K$ test then selects $K = \mathbb{Q}(i)$ uniquely.

| ID | Name (abbreviated) | Status | Regime / ref. | Layer |
|---|---|---|---|---|
| $H_{1,\mathrm{a}}$ | Type-II$_\infty$: static dS | Estab. | CLPW23 [?] | Hard core |
| $H_{1,\mathrm{b}}$ | Type-II$_\infty$: quasi-dS | Estab. | CP24; Speranza25 | Hard core |
| $H_{1,\mathrm{c}}$ | Type-II$_\infty$: FLRW asympt-dS | Estab. | KFLS24 [?] | Hard core |
| $H_{1,\mathrm{d}}$ | Type-II$_\infty$: 4D flat FRW $\xi$=1/6 | Estab. | ECI-int. [?] | Hard core |
| $H_{1,\mathrm{e}}$ | Type-II$_\infty$: gen. $\Lambda$CDM | Conj. | outside ECI | (not needed) |
| $H_{1,\mathrm{f}}$ | Type-II$_\infty$: Bianchi | Conj. | outside ECI | (not needed) |
| $H_{1,\mathrm{g}}$ | Type-II$_\infty$: massive/interact. | Conj. | outside ECI | (not needed) |
| $H_{1,\mathrm{h}}$ | Type-II$_\infty$: matter-$\Lambda$ bridge | Conj. | outside ECI | (not needed) |
| $H_2$ | KMS $\beta$=2$\pi$ (BW) | Estab. | | Hard core |
| $H_3$ | Microlocal funct. (PH$_k$ A6) | Estab. | | Hard core |
| $H_4$ | Structural inheritance | Estab. | | Hard core |
| $H'_5$ | CM-anchored attractor | W-Conf. | | Belt |
| $H_6$ | Hecke restriction $\mathcal{H}_1$ | Estab. | | Belt |
| $H'_7$ | Integer Damerell ladder | Conv. | A72 downgrade | Belt |
| $H'_8$ | Karam–Palatini Cassini wall | Post. | | Belt |
| $H'_9$ | KW dS-trap mechanism | Post. | | Belt |
| $H'_{10}$ | Leptogenesis $(n-1)^2$=6 | W-Conf. | | Belt |
| $H'_{11}$ | Option C (LYD20 retracted) | W-Conf. | | Belt |

Table 2: ECI v7.5 axiom set, post-A79 H1 partition (S5 graft 2026-05-06). H1 is now partitioned: 4 sub-classes Established ($H_{1,\mathrm{a}}$–$H_{1,\mathrm{d}}$; ECI exploits $H_{1,\mathrm{a}}$ and $H_{1,\mathrm{d}}$); 4 sub-classes Conjectural but outside the ECI Cassini-clean regime ($\xi = 1/6$, flat FRW, no Bianchi, no massive coupling) and NOT weakening the hard core. CLPW23 = arXiv:2206.10780; KFLS24 = arXiv:2406.01669 (A79 live-verified 2026-05-05). Estab. = Established; W-Conf. = Working-Confirmed; Conj. = Conjectural; Post. = Postulate. *[S5 coord. note: if S1 edits the H7' row above, parent must merge manually.]*

### 4.1 NPP20 lepton sector at $\tau_S = i$

The lepton sector follows Novichkov–Penedo–Petcov [?]: $S'_4$ doublet/triplet assignments for the three left-handed lepton generations, two heavy right-handed singlets, and the type-I seesaw with $\tau_S = i$ pinned by the residual $\mathbb{Z}_4 \subset \mathrm{SL}(2,\mathbb{Z})$ symmetry at the $S$-fixed point. The model has 4–6 free parameters for the 9 PMNS observables ($\theta_{12,13,23}$, $\delta_{CP}$, $\Delta m^2_{12,23}$, $\Sigma m_\nu$, $m_{\beta\beta}$, and one high-energy Majorana phase $\eta$). At $\tau_S = i$ the weight-2 $S'_4$ triplet form $Y_3^{(2)}$ acquires the alignment $(1,\ 1+\sqrt{6},\ 1-\sqrt{6})$ (DKLL19 Table 1, Case B, [?]), which feeds the CSD$(1+\sqrt{6})$ Littlest Modular Seesaw studied in Section **??**.

### 4.2 G1.12.B SU(5) modular $f^{ij}(\tau = i)$ campaign: M1–M2 PASS, M5 partial-width computation

The G1.12.B campaign [?] is the v7.4–7.5 proton-decay falsifier programme: a six-milestone modular SU(5) $5_H + 45_H$ construction targeting a parameter-free branching ratio

$$\frac{B(p\to e^+\pi^0)}{B(p\to K^+\bar{\nu})} \in [0.3,\ 3] \quad (95\%\ \mathrm{CL\ post\text{-}M_6}),$$

to be tested by Hyper-K + DUNE 2030+ data. The published v7.4 amendment recorded M1+M2 PASS; the present v7.5 amendment adds the M5 partial-width computation result. M3–M6 forecast is 3.5–5 sub-agent months at \$0–100.

**M1 (A22) PASS 4/4.** The 45-of-SU(5) Higgs is assigned to the $S'_4$ trivial singlet **1** at modular weight $k_{45} = 0$, with the off-diagonal Yukawa form-factor $f^{ij}(\tau)$ built from row-by-row LYD20-Model-VI contractions (rows $u, c, t$ at weights $1, 2, 5$, irreps $\hat{3}', \mathbf{3}, \hat{3}$). The $3\times 3$ matrix $f^{ij}(\tau = i)$

is computed symbolically (sympy) and numerically; all 6 off-diag entries are non-zero, modular invariance under $S$ and $T$ is verified to machine precision (residual $\leq 10^{-15}$), and the A2 +19.5% closure check matches at +19.43% (within 0.07 pp). The non-modular precedent is Haba–Nagano–Shimizu–Yamada [**?**].

**M2 (A26) PASS 5/5.** The off-diagonal up-quark Yukawa SVD at the W1 attractor $\tau^* = -0.1897 + 1.0034\,i$ reproduces the Antusch–Hinze–Saad GUT-scale targets [**?**] ($y_t = 0.4454$, $y_c/y_t = 2.725 \times 10^{-3}$, $y_u/y_c = 2.05 \times 10^{-3}$) to machine precision: the fit $(\beta_u/\alpha_u, \gamma_u/\alpha_u) = (152.42, 305.38)$ achieves $\chi^2_{\text{up}} = 6.7 \times 10^{-20}$ (2 parameters, 2 observables, exact); $|\Delta(y_c/y_t)| = 10^{-9}\%$. Unitarity of $U_L, U_R$ holds at $\sim 5 \times 10^{-16}$; SVD reconstruction at $\sim 7 \times 10^{-11}$.

**M5 partial widths (A36, this volume).** The full six-channel partial-width computation at the central $(M_{T_5}, M_{T_{45}}) = (10^{15}, 10^{12})$ GeV reference uses Haba eqs. (18)–(23) [**?**] (verified by direct PDF read), Aoki–Soni $\alpha_H(2\,\text{GeV}) = -0.0144\,\text{GeV}^3$ chiral parametrisation, FLAG-2024 lattice form factors [**?**], and the standard $D = 0.8$, $F = 0.46$, $f_\pi = 0.093\,\text{GeV}$, $A_{RL} = 2.6$, $\lambda_W = 0.22$ parameters. Six channels $\{p \to e^+\pi^0, \mu^+\pi^0, \bar{\nu}K^+, \mu^+K^0, e^+K^0, \bar{\nu}\pi^+\}$ are evaluated against current Super-Kamiokande limits [**?**, **?**] and the Hyper-K 20-yr [**?**] / DUNE 20-yr [**?**] reach. Numerical verification of all input constants is recorded in `notes/eci_v7_aspiration/A36_G112B_M5/tau_p_results.json`. The decisive falsifier remains unchanged from v7.4:

> *ECI v7.5 modular SU(5) $5_H + 45_H$ at $\tau \sim \tau^*$ (and quark-sector hierarchy at $\tau = i$ via dMVP26 Kähler) predicts $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ (95% CL post-$M_6$). Hyper-K + DUNE combined 2030+ data outside $[0.3, 3]$ at $> 3\sigma$, OR $\tau(p \to e^+\pi^0) > 3 \times 10^{35}$ yr at $> 2\sigma \Rightarrow$ **REFUTED**.*

The post-$M_6$ central forecast is $\tau(p \to e^+\pi^0) \sim 3 \times 10^{34}$ yr, $\tau(p \to K^+\bar{\nu}) \sim 10^{34}$ yr – both above current Super-K limits and well within Hyper-K + DUNE 20-yr reach.

**M23 audit: $B_{\text{Higgs}} = 1.077$, modular-geometry insight (v7.6 new).** Sub-agent M23 performed an independent amplitude computation (Path A) substituting the full modular Yukawa matrix $Y_{45}^{ij} = \kappa_i f^{ij}(\tau^*)$ into Haba–Nagano–Shimizu–Yamada eqs. (18)–(23) [**?**], bypassing the "2nd-generation-only" restriction of the Haba vanilla ansatz. At the central value $\kappa_u = 10^{-3}$:

$$A_\pi \;=\; 1.618 \times 10^{-5}, \quad A_K \;=\; 2.111 \times 10^{-5}, \quad B_{\text{Higgs}} \;=\; \frac{|A_\pi|^2 \cdot \text{PS}_\pi}{|A_K|^2 \cdot \text{PS}_K} \;=\; 1.077,$$

where the phase-space ratio $\text{PS}_\pi/\text{PS}_K = 1.8339$ uses PDG masses. *Key structural insight:* the dominant amplitude for $p \to K^+\bar{\nu}$ is the u-row / s-quark term ($A_K^{\text{u-row}} = 1.95 \times 10^{-5}$), 12.5× larger than the c-row term that Haba's vanilla ansatz uses. The ratio

$$\frac{B_{\text{Higgs}}}{1} \;\propto\; \left(\frac{f^{uu}}{f^{us}}\right)^2 \cdot \frac{\text{PS}_\pi}{\text{PS}_K}$$

is *independent of* $\kappa_u$ (it cancels), making $B_{\text{Higgs}} = 1.077$ a **modular-geometry prediction** fixed by the form-factor ratios $f^{uu}/f^{us}$ at $\tau = \tau^*$. The full B-ratio $B = B(e^+\pi^0)/B(K^+\bar{\nu})$ including gauge $X, Y$ exchange interference spans $[0.3, 5.5]$ depending on the relative phase (M23); the central estimate with canonical $M_{T_{45}} = 10^{14}$ GeV and constructive phase gives $B \approx 2.06$ (matching the v7.5 forecast), while the Higgs-only sub-ratio is 1.077. **Patel–Shukla prose fix:** the v7.5 text erroneously described arXiv:2310.16563 (Patel–Shukla 2023 [bibkey TBD, not yet entered]) as a proton-decay reference; the proton-decay amplitude derivation follows Haba [**?**]. Fine-tuning of $M_{T_{45}}$ corrected to 1/300–1/2000 of $M_{T_5}$.

### 4.3 CSD$(1+\sqrt{6})$ Littlest Modular Seesaw (A14)

A second probe of the $K = \mathbb{Q}(i)$ anchoring exploits the strict $S$-fixed point $\tau_S = i$ in the lepton sector. At $\tau_S = i$ the weight-2 $S_4'$ triplet form $Y_3^{(2)}$ admits the alignment

$$Y_3^{(2)}(i) \;\propto\; \big(1,\; 1+\sqrt{6},\; 1-\sqrt{6}\big), \tag{1}$$

the so-called CSD$(1+\sqrt{6})$ alignment of the Littlest Modular Seesaw [**?**, **?**]. Equation (**??**) drives a 2-real-parameter neutrino sector with two right-handed Majorana singlets, predicting normal ordering with $m_1 = 0$, Dirac CP phase $\delta_{CP} \approx -87^\circ$, first-octant $\sin^2\theta_{23} \approx 0.46$–$0.55$, and $\Sigma m_\nu \approx 0.06\,\text{eV}$. The cross-$K$ test of DKLL19 Table 1 (A14) confirms $K = \mathbb{Q}(i)$ uniqueness: only at $\tau_S = i$ ($K = \mathbb{Q}(i)$) does the weight-2 $S_4'$ triplet acquire the non-trivial nested-radical alignment $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$; at $\tau_{ST} = \omega$ ($K = \mathbb{Q}(\sqrt{-3})$) the alignment collapses to a permutation vector $(0, 1, 0)$, killing the seesaw.

**Falsifiers (Sec. ?? items 4–5).** DUNE 2030+ at $\pm 15^\circ$ on $\delta_{CP}$ rules out the prediction at $> 3\sigma$ if $\delta_{CP}$ centres at 0 or $+90^\circ$. JUNO 2030 NMO at 3–4$\sigma$ [**?**] independently kills the NO + $m_1 = 0$ prediction if inverted. KamLAND-Zen 2027 / nEXO 2030+ tests $m_{\beta\beta} \sim$ few meV.

### 4.4 $\sqrt{6}$ structure: A24 closure note

A PSLQ search at dps = 60, $|c| \le 500$, in seven distinct bases $\{\sqrt{6}, \Gamma(1/4), \pi, \log 2, \log 3, \Omega_K, \log\theta_{2,3,4}(i), \log\eta(i)\}$ (A24) yields no Q-linear relation of $\log\sqrt{6}$ with the Chowla–Selberg periods of $\mathbb{Q}(i)$: in every non-trivial relation found, the coefficient on $\log\sqrt{6}$ is zero. The $\sqrt{6}$ in Eq. (**??**) is therefore a Galois-rational quantity (it is the discriminant $\sqrt{b^2-4c}$ of the secular $2\times 2$ polynomial of $Y_3^{(2)}(i)$ eigenvalues), *not* a period-theoretic invariant of $\mathbb{Q}(i)$. Consequently the $K = \mathbb{Q}(i)$ anchor of CSD$(1+\sqrt{6})$ is *combinatorial* ($S$-fixed-point of $S_4'$), not a deep CM-period identity via Chowla–Selberg. This keeps the empirical falsifier (DUNE 2030+) valid but classifies the $1+\sqrt{6}$ alignment as a "thin coincidence" in the Lakatos sense, not a hard prediction lifted by deep number theory. Future probes: $\mathbb{Q}(\sqrt{-24})/\mathbb{Q}(\sqrt{-6})$ period rings (which contain $\sqrt{6}$ natively).

### 4.5 King–Wang multi-modulus dS-trap mechanism for $\tau = i, \omega$ (A47)

A14's empirical-deepness assessment ($P \sim 20\%$, "WEAK CONSTRAINT, deep") had flagged the absence of a physical motivation for $\tau_l = i$ from the modular-flavour literature. A47 closes this gap by grafting the King–Wang multi-modulus dS-trap of [**?**].

In the King–Wang setup, $\mathcal{N} = 1$ supergravity in the flavour modulus $\tau$ and dilaton $S$ has superpotential

$$W \;=\; \Lambda_W^3\, \Omega(S)\, \frac{H(\tau)}{\eta^6(\tau)}, \qquad H(\tau) \;=\; (j-1728)^{m/2}\, j^{n/3}\, P(j),$$

with $K \propto -\log(S+\bar{S})$ at tree level and Shenker-like $O(e^{-1/g_s})$ corrections. The key kinematic fact is that $\partial_\tau V$ has modular weight 2, so under the $S$-action at $\tau = i$ one has $(\partial_\tau V)|_i = (-i)^2(\partial_\tau V)|_i$, forcing $\partial_\tau V \equiv 0$ at $\tau = i$. The same logic via $ST$ (order 3) gives $\tau = \omega$. The Hessian is positive (KW Appendix C), so de Sitter minima emerge precisely at $\tau \in \{i, \omega\}$.

**Compatibility with ECI v7.5.** Three checks pass: (a) KW is purely modular-kinematic and makes no reference to CM points, Heegner values, or periods, so adding the ECI CM-by-$\mathbb{Q}(i)$ anchor is non-redundant. KW provides BOTH $i$ and $\omega$ vacua; ECI's A14 cross-$K$ test selects $\tau = i$ ($K = \mathbb{Q}(i)$) uniquely. (b) No double-counting: KW shapes $V(\tau, S)$, while the CM anchor selects *which* KW minimum is realised. (c) Cassini-clean: the KW $\tau$ is a SUGRA modulus with mass $m_\tau \sim m_{3/2}$ (TeV–$10^{12}$ GeV), whereas fifth-force/Cassini constrains scalars

with $m < 10^{-12}$ eV; this is a different scalar from the ECI NMC cosmological one and trivially evades the Cassini bound.

**Empirical-deepness upgrade.** A14's $P \sim 20\%$ ("WEAK, deep") is upgraded to $P \sim 35$–$40\%$ ("MEDIUM, deep") after the A47 graft, since the "WHY $\tau = i$?" question now has a string-cosmology answer (KW dS-trap) layered above the cross-$K$ selection. Honest deficits remain: KW supply no numerical $m_\tau$ or $\Lambda_W$, no CM/class-field-theory connection (ECI fills unilaterally), and a Shenker-type uplift rather than KKLT-tuned dS (no strict no-tuning theorem).

### 4.6 Leptogenesis from CSD$(1+\sqrt{6})$ at $\tau_S = i$: the $(n-1)^2 = 6$ structural signature (A55)

The A14 graft (Section **??**) places the lepton sector at the modular fixed point $\tau_S = i$ of $S_4'$, where the weight-2 triplet alignment $(1,\, 1+\sqrt{6},\, 1-\sqrt{6})$ yields the CSD$(1+\sqrt{6})$ Littlest Seesaw of King [**?**, **?**] with two right-handed neutrinos and $m_1 = 0$. This subsection records the implied baryon asymmetry of the universe (BAU) via flavoured thermal leptogenesis.

For CSD$(n)$ Case A ($M_{\rm atm} = M_1 \ll M_{\rm sol} = M_2$), the hierarchical-limit decay-asymmetry parameters [**?**] read

$$\epsilon^A_{1,\mu} \simeq -\frac{3}{16\pi}\frac{M_1}{M_2}\, n(n-1)\, b^2 \sin\eta, \tag{2}$$

$$\epsilon^A_{1,\tau} \simeq -\frac{3}{16\pi}\frac{M_1}{M_2}\, (n-1)(n-2)\, b^2 \sin\eta, \tag{3}$$

whose flavour-summed total scales analytically as

$$\epsilon^A_{1,\rm tot} \;=\; -\frac{3}{16\pi}\frac{M_1}{M_2}\, 2(n-1)^2\, b^2 \sin\eta. \tag{4}$$

Substituting the ECI value $n = 1+\sqrt{6}$ from the modular fixed-point gives the exact rational signature

$$(n-1)^2 \;=\; (\sqrt{6})^2 \;=\; 6, \qquad \Longrightarrow \qquad \boxed{\frac{Y_B^{\rm CSD(1+\sqrt{6})}}{Y_B^{\rm CSD(3)}} \;=\; \frac{(n-1)^2}{(3-1)^2} \;=\; \frac{6}{4} \;=\; \frac{3}{2}}, \tag{5}$$

at fixed Yukawa amplitude $b$, high-energy Majorana phase $\eta$, and heavy Majorana masses $M_{1,2}$. The irrationality of $n = 1+\sqrt{6}$ *cancels* in $(n-1)^2$, leaving an integer factor 6 that mirrors the 3 in CSD(3) – a structural fingerprint of the modular $\tau_S = i$ alignment.

Calibrating to the King–Molina-Sedgwick–Rowley benchmark Case A2 ($M_1 = 5.05\times10^{10}$ GeV, $M_2 = 5.07\times10^{13}$ GeV, $a = 0.00806$, $b = 0.0830$, $\eta = 2\pi/3$), which gives $Y_B^{\rm CSD(3)} = 0.860\times10^{-10}$ at $\chi^2/{\rm dof} = 1.51/3$, Eq. (**??**) predicts

$$Y_B^{\rm CSD(1+\sqrt{6})}\Big|_{a,b,\eta,M_R\ \rm fixed} \;=\; 1.290\times10^{-10}, \tag{6}$$

which exceeds the Planck value $Y_B^{\rm obs} = (0.872 \pm 0.006)\times10^{-10}$ [**?**] by 48%. This is reabsorbed exactly by the rescaling $b^2\sin\eta \to (2/3)\, b^2 \sin\eta$ (e.g. $b \to 0.0678$ at $\eta = 2\pi/3$), which leaves the low-energy PMNS parameters $(\theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, \Delta m^2_{12}, \Delta m^2_{13})$ shifted by $\lesssim 1\sigma$ relative to the King Case A2 benchmark, well within the NuFIT 5.3 envelope. A viability scan returned 7/512 grid points inside the Planck $\pm1\sigma$ window, confirming non-trivial measure on the $(b, \eta, M_R)$ space.

**Falsifier status.** ECI v7.5 does not promise a *parameter-free* BAU prediction in the A14/CSD($1+\sqrt{6}$) sector – the high-energy phase $\eta$ is independent of the low-energy Dirac phase $\delta_{CP} \approx -87°$ in any two-RHN seesaw with one phase [**?**]. What v7.5 *does* predict is the structural signature $(n-1)^2 = 6$ in $\epsilon_1$ scaling, testable indirectly via the joint $(\theta_{12}, \theta_{13}, \theta_{23}, \delta_{CP}, m_2, m_3, Y_B)$ $\chi^2$/dof fit on DUNE+JUNO+NuFIT 2030+ data. Honest probability assessment: graft into v7.5 successfully $\sim 55\%$, sharp falsifier added $\sim 10\%$, Foundational-Prize-class $\sim 3\%$.

### 4.7 $N = p^2$ interlinking: structural unity of M13's three arithmetic findings (M22, v7.6 new)

Sub-agent M22 ($\beta$-renormalized monotonicity validation, 2026-05-06) discovered a structural meta-finding that unifies the three independent arithmetic facts of the M13 campaign for $f_{4.5.b.a}$ (LMFDB 4.5.b.a, weight 5, CM by $\mathbb{Q}(i)$, level $N = 4 = 2^2$):

1. **Finding 1 (Steinberg-edge classical condition, M20 reframe):** The identity $|a_p| = p^{(k-1)/2}$ at the ramified prime $p = 2$ gives $a_2 = -4$ (M20 live-verified via LMFDB direct fetch; 8 weight-5 CM newforms tested, 8/8 MATCH). M20 showed this is the *classical critical-slope condition* of Shimura (1971) / Ribet (1977): $\alpha_p = \beta_p = p^{(k-1)/2}$ forces Euler factor $(1 + p^{(k-1)/2}T)^2$, making the Pollack $\pm$-split degenerate. Classical, NOT a novel ECI result.

2. **Finding 2 (functional-equation symmetrization, M21):** At the Steinberg edge, $\alpha_2 + \alpha_3 = 1/8$ holds for 4.5.b.a (three independent conditions converging).

3. **Finding 3 ($\beta$-renormalized monotone sequence, M22):** The F1 renormalization $\alpha_m \cdot (-2^{m-1}) \cdot (1+2^{m-3})$ yields $v_2 = \{-3, -2, 0, +1\}$, the *unique* strictly monotone pattern among 8 formulas tested (F2–F8 give ties, oscillations, or degeneracies). The $(1 + p^{m-3})$ factor arises from local Euler-factor structure at $\beta = 0$ (F1 is theoretically motivated, not ad hoc).

**$N = p^2$ meta-finding.** M22's cross-newform check reveals that the rationality of $\alpha_3/\alpha_2$ (required for $v_2 \in \mathbb{Z}$) demands $N = p^2$. For small CM weight-5 newforms: 4.5.b.a ($N = 4 = 2^2$): $v_2$ well-defined, all 3 findings hold; 7.5.b.a ($N = 7$ prime): $\alpha_3/\alpha_2$ irrational, $v_2$ undefined; 8.5.d.a ($N = 8 = 2^3$): same obstruction; 12.5.c.a ($N = 12$): same. **Conclusion:** 4.5.b.a is structurally special precisely because $N = 4 = 2^2$ is a perfect square, a consequence of $K = \mathbb{Q}(i)$ having discriminant $d_K = -4 = -2^2$. All three M13 findings are interlinked via this single algebraic fact.

The paper-2 (math.NT, see §**??**) must derive F1 from Frobenius-degeneracy compensation BEFORE claiming monotonicity as evidence for $p$-adic interpolation (M22 recommendation).

## 5 Quark hierarchy via dMVP26 canonical Kähler at $\tau = i$ (A48 partial graft)

The G1.12.B SU(5) framework (Section **??**) supplies modular structure but consumes hierarchies. We graft de Medeiros Varzielas–Paiva [**?**] (`arXiv:2604.01422`) to supply the quark-mass hierarchy at $\tau = i$.

With $S_4'$ weights $(k_Q, k_{u^c}, k_{c^c}, k_{t^c}, k_{d^c}, k_{b^c}) = (0, -6, -1, -10, -6, \ldots)$ (compatible with row-by-row LYD20 contractions of Section **??**) and Kähler metric

$$K \supset -\Lambda_K^2 \log[-i(\tau - \bar{\tau})] + \sum_i \frac{|\phi_i|^2}{[-i(\tau - \bar{\tau})]^{k_i}},$$

canonical normalisation absorbs $(2\,\mathrm{Im}\,\tau)^{-k_i/2}$ into the up-quark mass matrix $M_u$ and down-quark mass matrix $M_d$, giving $y_u/y_c,\ y_c/y_t \sim 10^{-3}$ with real $\mathcal{O}(1)$ Yukawas $C_i \in [0.4, 5.6]$.

**Survival at $\tau = i$ (A48 numerical check).** For ECI v7.5 we PIN $\tau = i$ (CM-anchor at $K = \mathbb{Q}(i)$). The authors of [?] prove on p. 9 via analytical expansion $\tau = i + \epsilon$ that $J_{CP} \propto \mathrm{Re}(\epsilon)$ "completely decoupling the CP phase from the mass hierarchies, which are primarily related to $\mathrm{Im}\,\tau \sim 1$". The A48 SVD at $\tau = i$ versus the dMVP26 best fit $\tau^{\mathrm{dMVP26}} = 0.00455 + 1.00705\,i$ confirms shifts $< 5\%$ in the up-sector and $\sim 20\%$ in $y_d/y_s$:

| Ratio | PDG-2024 (GUT) | dMVP26 best-fit | $\tau = i$ pinned (A48) |
|---|---|---|---|
| $y_c/y_t$ | $2.81 \times 10^{-3}$ | $2.99 \times 10^{-3}$ | $3.09 \times 10^{-3}$ |
| $y_d/y_s$ | $5.00 \times 10^{-2}$ | $2.27 \times 10^{-1\,*}$ | $1.81 \times 10^{-1\,*}$ |
| $y_s/y_b$ | $1.78 \times 10^{-2}$ | $1.78 \times 10^{-2}$ | $1.86 \times 10^{-2}$ |

Table 3: A48 SVD test of dMVP26 hierarchies under $\tau = i$ pinning (asterisk: a residual normalisation gap of order $\times 4$ in $y_d/y_s$ relative to PDG affects both the best fit and the $\tau = i$ point and is traced to a likely Eq. (24) Clebsch–Gordan sign or weight-1 $\hat{Y}_3$ normalisation; the $\tau = i$/best-fit *ratio* is the relevant graft viability indicator and is $< 5\%$ in the up sector and $\sim 20\%$ in $y_d/y_s$). Authors' own analytic argument $J_{CP} \propto \mathrm{Re}(\epsilon)$ is the independent confirmation.

**Cost: $J^q_{CP} = 0$ at strict $\tau = i$.** At $\mathrm{Re}(\tau) = 0$ the CKM Jarlskog vanishes, so the CP-phase of CKM cannot come from the modular sector at strict $\tau = i$. v7.5 lists two compensator routes (open):

- *(a) Lepton-sector spurion:* the $\delta_{CP} \approx -87^\circ$ of $\mathrm{CSD}(1 + \sqrt{6})$ (Section **??**) feeds the CKM phase via a GUT-scale Yukawa-mixing term.
- *(b) RGE threshold* $\mathrm{Re}\,\tau \sim 10^{-3}$*:* a small renormalisation-group displacement of $\tau$ from $i$ at the GUT-matching scale, of order $\mathrm{Re}\,\tau \sim 10^{-3}$, would give the observed Jarlskog $J^q_{CP} \sim 3 \times 10^{-5}$ via the linear scaling of [?] p. 9. This compensator is not yet derived from G1.12.B M3–M4 RGE matching; it is registered as a quantitative open question in Section **??**.

Both routes are compatible with the v7.5 Option-C architecture; neither is yet a parameter-free derivation. The A48 graft is therefore PARTIAL: hierarchies survive, CP phase is cost.

# 6 ECI production posterior on real DESI/Pantheon+/Planck data (A71 + M24, v7.6 new)

## 6.1 Pipeline and convergence

Sub-agent A71 ran the first ECI NUTS production posterior on real cosmological data (DESI DR1 BAO + Pantheon+ SNeIa + Planck 2018 CMB). The pipeline uses `cosmopower-jax` (386 predictions/s on the RTX 5060 Ti), the JAX named_shape patch, and `numpyro` NUTS sampler. Run configuration:

- 4 chains $\times$ 4,000 warmup + 8,000 samples; total 1133 s wall time.
- Convergence: $\hat{R}_{\max} = 1.0001$; $\mathrm{ESS}_{\min} = 30{,}683$ (both excellent).
- Likelihood: M1-corrected (2 likelihood bugs fixed by M1).
- $\theta_{\mathrm{MC}}$ approximation: real $\theta_{\mathrm{MC}}$ not yet computed; preliminary diagnostic shows $3.1\sigma$ offset vs $\Lambda$CDM – acceptable for smoke test.

### 6.2 Posterior marginals

| Parameter | ECI posterior (mean $\pm 1\sigma$) | Note |
|---|---|---|
| $H_0$ | $68.51 \pm 0.14$ km/s/Mpc | Planck-like; vs LCDM-only 71.83 |
| $\omega_b$ | $0.02443 \pm$ [TBD] | Planck-consistent |
| $\omega_c$ | $0.11898 \pm$ [TBD] | |
| $n_s$ | $0.96492 \pm$ [TBD] | |
| $\log_{10} A_s$ | $3.044 \pm$ [TBD] | |
| $\tau$ | $0.054 \pm$ [TBD] | |
| $\xi_\chi$ | $-0.0240 \pm 1.8 \times 10^{-5}$ | RAIL FINDING (see below) |

The $H_0 = 68.51 \pm 0.14$ value is Planck-like and significantly below the SH0ES-leaning $H_0 = 71.83$ that a $\Lambda$CDM-only run on the same dataset would yield. This is consistent with ECI's NMC sector pulling $H_0$ toward the Planck anchor via the $\xi_\chi$-dependent effective Newton constant $G_{\rm eff}(z)$.

### 6.3 $\xi_\chi$ rail diagnostic (M24)

The $\xi_\chi$ posterior rails at $-0.0240 \pm 1.8 \times 10^{-5}$: the mean is 14× the standard deviation from zero, and the posterior is extremely tight relative to the prior. Sub-agent M24 ran a diagnostic with a widened prior $[-0.10, +0.10]$ (vs the original $[-0.024 \pm \text{tight}]$) and found the chain still rails at $-0.0240 \pm 2 \times 10^{-5}$. This confirms that $\xi_\chi = -0.024$ is a **real data signal**, NOT a prior artefact.

**Interpretation via RG running (A73).** Sub-agent A73 computed the RG running of the NMC coupling in the ECI framework and found $\xi(M_Z) \simeq +0.001$, evolving to $\xi(M_{\rm GUT}) \simeq -0.029$ under $\sim 14$ decades of running. The cosmological "effective $\xi$" sensed by CMB + LSS data, $\xi_\chi^{\rm eff} \simeq -0.024$, is *consistent with the RG trajectory*: the UV value $\xi_Z = +0.001$ and IR cosmological value $\xi_{\rm eff} = -0.024$ are both within the A73 RG-stable wedge $\xi \in [-0.029, +0.001]$. This is NOT a falsification of ECI; it is a *success*: RG running from $M_Z$ to $M_{\rm GUT}$ changes the sign of $\xi$, and the cosmological probe is sensitive to the GUT-scale effective value.

**Caveats and v7.7 refinements.**

- $\theta_{\rm MC}$ **approximation:** a real $\theta_{\rm MC}$ solver will be integrated in v7.7 (queue: replace approximate $\theta_{\rm MC}$ with CAMB-accurate computation).
- **Planck off-diagonal covariance:** the current run uses diagonal CMB likelihood; full off-diagonal covariance matrix will tighten $\xi_\chi$ constraint in v7.7.
- **Wolf–KG production run:** a parallel run with the Wolf 2025 $\xi_\chi$ prior ($\xi \in [+1, +4]$) is queued for the three-way Bayes contest (§**??**).

## 7 Two-tau MUDDY note: empirical helpfulness vs. GUT-unification preference (A42 + A49 correction)

The v7.4 strategic synthesis (commit `6.0.53.5`) included a row in A42's TOP-5 BSM comparison claiming that "two-$\tau$ models are 60× worse than single-$\tau$". The A49 autopsy of Du–Wang [**?**] (`arXiv:2209.08796`, PDF directly read) corrects this as a category error: the "60×" arose from comparing the single-$\tau$ Sample IX′ sweet spot (Scenario III, $\chi^2_{\rm tot} = 1.58$) against Scenario III's generic-bad sub-scenarios in dual-$\tau$ ($\chi^2 \approx 95$, different sub-models). The correct table-by-table comparison is:

| Sub-scenario (lepton sector only) | Single $\tau$ | Two $\tau$ |
|---|---|---|
| Scenario I (Du–Wang) | $\chi_l^2 = 236.3$ | $\chi_l^2 = 48.8$ |
| Sub-scenario IX (Scenario II) | $\chi_l^2 = 486.0$ | $\chi_l^2 = 30.2$ |
| Sub-scenario X′ (Scenario III) | $\chi_l^2 = 233.0$ | $\chi_l^2 = 58.9$ |
| Sample IX′ (Scenario III sweet spot) | $\boldsymbol{\chi^2_{\text{tot}} = 1.58}$ | not reported |

Table 4: A49 honest re-tabulation of Du–Wang Tab. 5: two-$\tau$ *improves* the lepton fit by 4–16× in every sub-scenario where both numbers appear. Sample IX′ wins because flipped SU(5) GUT-locks $Y^D = (Y^E)^T$ (one Yukawa makes both down-quark and charged-lepton matrices), so the lepton sector inherits any tuning that succeeds for quarks, plus the $(1', 1'', 1'')$ singlet assignments with weights $(2, 4, 2)$ for $\bar{F}_{1,2,3}$ provide enough freedom for one $\tau$ in the fundamental domain to satisfy quark hierarchy AND PMNS angles simultaneously. Mechanism is GUT-locked structure + lucky weight assignment, NOT cancellation.

**Du–Wang's own theoretical objection to two-$\tau$.** Du–Wang [**?**, §4, conclusions] themselves argue that two-$\tau$ is theoretically problematic: GUT representations (flipped SU(5) $\bar{10}$, SO(10) **16**) contain BOTH quarks and leptons, so independent $\tau_q, \tau_l$ "spoils unification of matter contents". They propose a 5D $M_4 \times S^1/Z_2$ orbifold that breaks $A_4^Q \times A_4^L \to$ diagonal $A_4^D$ via boundary conditions (their eqs. 4.13–4.16), restoring a single low-energy modular symmetry.

**v7.5 reformulation of the single-$\tau$ axiom $H_5'$.** The A49 correction obliges us to drop the *empirical* "two-$\tau$ is 60× worse" argument for single-$\tau$ and replace it by a STRUCTURAL one:

> *The v7.5 single-$\tau$ choice is justified on UV matter-unification grounds (Du–Wang orbifold-BC reduction of $A_4^Q \times A_4^L \to$ diagonal $A_4^D$, generalised to $S_4'$). The CM-by-$\mathbb{Q}(i)$ anchor selects $\tau_S = i$ as the high-symmetry IR fixed point at which one modulus suffices. If v7.5 wants UV two-$\tau$, the orbifold-BC reduction becomes a predicted constraint, not freedom.*

The A16 caveat of v7.4 (single-$\tau$ refit pulls $\sin^2\theta_{13}$ at $24.5\sigma$ under the LYD20 unified-scaffold embedding) is now acknowledged as confirmation of the A46 retraction: it was the LYD20 unified-scaffold embedding that broke under single-$\tau$ pinning, not single-$\tau$ itself. The v7.5 Option-C architecture (NPP20 lepton + G1.12.B quark) does not exhibit the A16 pull because NPP20 has independent lepton-sector freedom.

# 8 Sub-algebra $H_1$ closure, integer-point Damerell ladder, and the A17 numerological CKM hits (downgraded per A72)

The arithmetic core of v7.4 – the integer-point Damerell ladder $H_7'$ for $f_{4.5.\text{b.a}}$ at $K = \mathbb{Q}(i)$ – is preserved in v7.5 as an *arithmetic anchor* but is **downgraded** from a structural prediction tool to a **notation convention** following the A72 null test (Section **??**). We summarise the result for self-containment.

**Theorem 1** (Bernoulli-anchored Cardy hit, v7.4 Theorem 2 reused)**.** *For the LMFDB CM newform $f_{4.5.b.a}$ (CM by $\mathbb{Q}(i)$, weight $k = 5$, level $N = 4$) and the Chowla–Selberg period $\Omega_K = \Gamma(1/4)^2/(2\sqrt{2\pi}) \approx 2.62205755$, the algebraic part $\alpha_2 := L(f_{4.5.b.a}, 2)\,\pi^2/\Omega_K^4$ equals $\alpha_2 = 1/12 = B_2/2 = -\zeta(-1)$, yielding the Cardy density $\rho_{\text{Cardy}}(c = 1) = c/12 = 1/12$ of the free boson parameter-free. The $\Gamma$-factor functional equation forces $\alpha_3 = \Gamma(2)/\Gamma(3) \cdot \alpha_2 = 1/24$, reproducing the $c = 1/2$ Ising central charge as a structural $\Gamma$-shadow. The full ladder $\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} = \{1/10, 1/12, 1/24, 1/60\}$ is the standard Eisenstein–Kronecker ladder for a weight-5 CM-by-$\mathbb{Q}(i)$ newform [**?**, **?**, **?**, **?**, **?**, **?**].*

**A72 downgrade of $H_7'$.** The Damerell-ladder rationals $\{\alpha_m\}$ characterise the CM $L$-values of $f_{4.5.b.a}$ at $K = \mathbb{Q}(i)$ and are *mathematically* rigorous (Hurwitz/Chowla-Selberg/Damerell/Shimura anchoring). However, the A72 null test (Section **??**) establishes that the $\alpha_m$ ladder has **no detectable structural signal on low-energy fermion observables**: the full Damerell-derivable invariant space (150 candidates from products $\alpha_i \cdot \alpha_j$, $\alpha \cdot \pi^k$, $\alpha \cdot \sqrt{n}$, $w_K$ factors) compared against 19 PDG-2024/NuFit-5.3 PMNS+lepton+quark observables yields $Q(i) = 358$ *below* the random-rational null mean $366 \pm 25$ ($\sigma = -0.33$, $P_{\text{null}} = 0.63$, strict cut $\sigma_{\text{hit}} < 0.3$, $|q| \leq 1$). Accordingly, $H_7'$ is reclassified as a **notation convention** for the Damerell-derived rationals characterising CM $L$-values at $K = \mathbb{Q}(i)$ (LMFDB 4.5.b.a, Hecke $H_1$, $\chi_4$ nebentypus structure), **not** a falsifiable structural input for low-energy fermion observables (cf. A72, `null_test_results.json`). The Cardy hits ($\alpha_2 = 1/12$, $\alpha_3 = 1/24$) are *mathematical* facts about $f_{4.5.b.a}$ that survive A72 unchanged; what A72 kills is the claim that the same ladder *predicts* CKM/PMNS values beyond random expectation.

**A17 numerological CKM hits (demoted to speculative observations, A62+A72).** The two A17 alignments of v7.4 are retained for audit trail as speculative numerical observations only ($P_{\text{null}} > 30\%$, A72; $< 1.7\sigma$ vs random, A62):

- $|V_{us}| = (9/4)\,\alpha_1 = 9/40 = 0.22500$, matching PDG 2024 $0.22501 \pm 0.00068$ to $0.015\sigma$ — **speculative numerical coincidence, NOT a structural prediction** (A62: CKM-Damerell null test $< 1.7\sigma$ vs random; A72: full-space $\sigma = -0.33$, $P = 0.63$). *Do not cite as an ECI prediction.* No statistical preference vs random rational ladders (A72).
- $|V_{cb}|^2 = \alpha_1\,\alpha_4 = 1/600 \approx 1.667 \times 10^{-3}$ — **RETIRED by A62**: HFLAV-2025 average $|V_{cb}| = 0.04183(79)$ places $1/600$ at $1.26\sigma$ (was $< 0.1\sigma$ on HFLAV-2024 used in v7.4). Retained for audit trail only.

The multiplicative coefficients $(9/4, 1)$ remain flagged *empirical*. Cross-$K$ confirms $K = \mathbb{Q}(i)$ Bernoulli/Hurwitz uniqueness for the *mathematical* ladder, but this is a tautology (different $K$ = different Hurwitz rationals) carrying no independent SM evidence. A72 closes the question.

**What survives.** The LMFDB 4.5.b.a CM newform anchor, Hecke $H_1$ restriction, and $\chi_4$ nebentypus structure (Section **??**) are *mathematical anchors* validated by P-NT Galois descent and are unaffected by A72. Explicitly: $K = \mathbb{Q}(i)$ is the arithmetic anchor of $f_{4.5.b.a}$; numerology of low-energy fermion observables built from its Damerell ladder does not exceed random.

The unfit central charges $c = 7/10$ (tricritical Ising) and $c = 4/5$ (tetracritical Potts) of the unitary minimal series remain open: either they arise from the BDP $p$-adic anti-cyclotomic complement at $s = k/2$, or they require a second CM newform (the candidate is LMFDB 16.5.c.a, level 16, $\chi_4$ nebentypus, paired with $f_{4.5.b.a}$ in P-NT companion identifications of hatted weight-5 multiplets). Discriminating between these two options is the next concrete arithmetic test of v7.5.

**Brown–Fonseca single-valued periods.** The $\Omega$-independent ratio $R(f) = 6/5$ is a natural candidate for a single-valued period in the sense of Brown–Fonseca [**?**], who interpret the Gross–Zagier conjecture via motivic biextensions $\mathrm{Sym}^k H^1(E)$ on moduli stacks $M_{1,n}$. Extending their geometric proof (level 1, weight 4, mixed Tate) to level 4, weight 5 is an open problem requiring that $M_{1,3}^{\Gamma_1(4)}$ be mixed Tate over $\mathbb{Z}[i, 1/2]$. The candidate biextension is $B_f = \mathrm{Sym}^4 H^1(E_i) \otimes \mathbb{Q}(\chi_{-4})$ for $E_i = \mathbb{C}/\mathbb{Z}[i]$.

# 9 Conditional analytical $\tau$ derivations: two $H_6$-co-dependent partial chains (A54)

Two analytical chains have been explored to derive the $\tau_S = i$ fixed point from arithmetic principles. Both are *partial* and $H_6$*-co-dependent* – they do not constitute "two independent analytical derivations". The v7.5 amendment records this honestly per A54 scoping.

**Chain A: Galois descent on the $\Gamma(4)$ modular curve (P-NT).** A bijective map between hatted weight-5 $S_4'$ multiplets and a pair ($f_{4.5.\mathrm{b.a}}$, $f_{16.5.c.a}$) of LMFDB CM newforms over $\mathbb{Q}(i)$ [**?**] forces $\tau_S = i$ via the unique $\Gamma_0(4)$-fixed point of the Galois action; the $\chi_4$ nebentypus restriction $H_6$ selects $\alpha_2 = 1/12 = B_2/2$ as the canonical algebraic part. This is the chain written up in the companion P-NT paper (BLMS submission queue).

**Chain B: Bost–Connes $\times$ CM at $\beta = 2\pi$ (A54 scoping).** A new chain uses the Connes–Marcolli–Ramachandran KMS-states-$\times$-complex-multiplication framework [**?**] (`arXiv:math/0501424`) for imaginary quadratic $K$ including $K = \mathbb{Q}(i)$. The five-step chain is: (S1) BC$\times$CM construction admits a $\beta = 2\pi$ formulation (✓ via CMR05); (S2) the 2-fold extremal-KMS ambiguity $i$ vs. $-i$ for $h(\mathbb{Q}(i)) = 1$ (open); (S3, dominant obstruction) $\beta = 2\pi$ does NOT pick out infinity-type $(4, 0)$; the $H_6$ $\chi_4$ nebentypus is still required as input – Chain B does *not* improve on Chain A's $H_6$-dependency; (S4) the type-$\mathrm{III}_1/\mathrm{II}_\infty$ obstruction reappears when identifying $\beta = 2\pi$ between the ECI-side DEHK [**?**] type-$\mathrm{II}_\infty$ algebra and the CMR05 type-$\mathrm{III}_1$ algebra of BC$\times$CM (the bypass of A43 [**?**] is only PARTIAL); (S5) the Grössencharakter $\psi \to f_{4.5.\mathrm{b.a}}$ $\theta$-lift via Hecke 1937 (✓).

**Honest verdict.** Chain B is genuinely new and publishable as a conditional math-ph note in 6–8 weeks sub-agent budget; full closure as an analytical theorem requires 24+ months and a co-author (Marcolli, Neshveyev). The corrected probability of closure as a clean derivation of $\tau = i$ *independent of* $H_6$ is $\sim 12\%$, down from the earlier Opus scoping estimate of 25%. v7.5 therefore records the chain as a SECOND $H_6$-co-dependent partial derivation, not a second independent one.

**Brief-internal correction (logged not propagated).** The earlier internal scoping had cited `arXiv:hep-th/9407124` as "Bost–Connes 1995" – this ID is in fact Gelfand–Krob–Lascoux–Leclerc–Retakh–Thibon "Noncommutative symmetric functions". The actual Bost–Connes 1995 paper (Selecta Math. NS 1.3, 411–457) is pre-arXiv. The chain's primary reference is CMR05 (`arXiv:math/0501424`), which generalises BC$\times$CM to imaginary quadratic $K$ including $\mathbb{Q}(i)$. No fabricated bibliography entry was produced; the brief-internal mis-tag is logged in A54 SUMMARY (hallu counter unaffected).

# 10 Wolf–ECI–AKW three-way NMC contest (M9 reframe, v7.6 new)

## 10.1 Framework

Three NMC models are now in active competition for the $\xi_\chi$ cosmological signal:

1. **Wolf 2025** ([**?**]): $\xi_\chi = 2.31 \pm 0.44$ (large positive NMC, King–Gower test FAILING by M7 audit).

2. **ECI v7.5/7.6** (this work): $\xi_\chi = -0.0240$ (small negative NMC, RG-consistent, King–Gower passing).

3. **AKW 2026** (`arXiv:2604.08449`, Antusch–King–Wang 2026 [unverified-this-session; bibkey AKW26 not yet entered]): $\xi_\chi \approx 0$ class (modular flavour + NMC at the Cassini wall).

Sub-agent M9 (reframing session, 2026-05-06) established that ECI and AKW belong to the *same class*: both are $\xi \approx 0$ King–Gower passing models with modular-flavour anchors. Wolf 2025 is **disqualified** from this class because M7 (Wolf–NMC–KG ODE audit) found 4/4 independent SMOKE-PASS tests of the Wolf ODE structure, but the combination of large $\xi = 2.31$ with the Cassini bound analytically forces $(\chi_0/M_P)^2 < 3 \times 10^{-6}$ (KSTD 2026, Section **??**), which Wolf 2025 circumvents only via a Palatini sub-branch.

**King–Gower (KG) gate.** A model passes the KG gate [**?**] if its NMC sector generates an observationally consistent $G_{\rm eff}(z)$ evolution without violating Solar-System Cassini bounds. ECI metric-branch: PASS (Cassini wall structurally enforced, $\xi_\chi = -0.024 \ll 6 \times 10^{-6}$ wall for $\chi_0/M_P \sim 1$). Wolf 2025: FAIL in metric branch (KG contradiction at $\xi = 2.31$); CONDITIONAL PASS in Palatini sub-branch only.

### 10.2 Strategic options

**Option A (recommended):** maintain a 2-way KG-gate paper (ECI vs ΛCDM), cite AKW (`arXiv:2604.08449`, [unverified]) in §2 as a distinct $\xi \approx 0$ model in the same passing class, defer 3-way Bayes contest to production run (v7.7).

**Option B (future):** 3-way Bayes contest $\ln B_{\rm ECI/Wolf}$ vs $\ln B_{\rm AKW/Wolf}$ once Wolf–KG production run completes.

### 10.3 Top-5 degeneracy-breaking observables

The following five probes distinguish ECI ($\xi \approx -0.024$) from Wolf ($\xi = +2.31$) and from AKW ($\xi \approx 0$):

1. $G_{\rm eff}(z)$ growth-rate evolution (DESI Y3 + LSST 2030): ECI predicts small $G_{\rm eff}$ deviation from $G_N$; Wolf predicts $G_{\rm eff}(0)/G_N \in [1.4, 1.8]$.

2. $f\sigma_8(z)$ (Euclid 2027+): ECI $f\sigma_8$ close to ΛCDM; Wolf $f\sigma_8$ enhanced by large-$\xi$ boost.

3. $w(z)$ shape (DESI Y5 + Rubin): both ECI and Wolf have $w \approx -1$ at background level; $\xi$-dependent perturbation structure differs.

4. $m_{\rm DM}(z)$ (effective dark-matter mass from NMC coupling): ECI negligible; Wolf non-negligible.

5. CMB B-modes: ECI predicts $r < 10^{-6}$ (modular $\tau$-inflation, M18 correction of M9); Wolf–2025 does not specify $r$. If LiteBIRD 2029+ detects $r > 10^{-3}$, ECI modular $\tau$-inflation is **REFUTED**.

**$r$-prediction (M18 corrects M9).** Sub-agent M18 found that M9's original ECI $r > 0$ estimate was an arithmetic error. The corrected ECI modular $\tau$-inflation prediction is $r < 10^{-6}$ (well below LiteBIRD's $r \sim 10^{-3}$ threshold). **If LiteBIRD detects $r > 10^{-3}$, ECI modular $\tau$-inflation is REFUTED** and the inflation sector must be replaced. The NMC cosmological sector is not affected.

# 11 Adelic Katz paper-2: Conjectural framework (M3+M13+M20+M22, v7.6 new)

## 11.1 Background: failure of Pollack± and classical rescue

The CM newform $f_{4.5.b.a}$ (LMFDB 4.5.b.a, weight $k = 5$, level $N = 4$, CM by $\mathbb{Q}(i)$, nebentypus $\chi_4$) has $a_2 = -4$, placing it on the *critical slope* at $p = 2$: $v_p(a_p) = (k-1)/2 = 2$. The standard Pollack ±-construction (Pollack 2003, *Int. Math. Res. Not.* **2003**(33); [bibkey TBD, not yet entered]) requires $v_p(a_p) < (k-1)/2$, so it **fails** for $f_{4.5.b.a}$.

M20 (LMFDB cross-check, 2026-05-06) confirmed that this is the classical Steinberg-edge / critical-slope condition: 8/8 weight-5 CM newforms tested satisfy $|a_p| = p^{(k-1)/2}$ at ramified primes (with $\psi$ unramified at $\pi$), universally. The explanation is classical: $a_p = \psi(\pi) \cdot p^{(k-1)/2}$, and for a Hecke character unramified at $\pi$, $|\psi(\pi)| = 1$, so $|a_p| = p^{(k-1)/2}$. This was known to Shimura (1971) and exploited by Ribet (1977); it is **not novel**.

## 11.2 Functional-equation symmetrization (M21)

At the Steinberg edge, the standard functional equation $\Lambda(f, s) = \epsilon_f N^{s-k/2} \Lambda(f, k-s)$ with $N = 4$, $k = 5$, $\epsilon_f = 1$ for 4.5.b.a gives a clean symmetrization. M21 found the specific relation $\alpha_2 + \alpha_3 = 1/8$ for this form under three convergent conditions, establishing that the functional equation is particularly clean for 4.5.b.a. This is **specific** to the $N = p^2 = 4$ structure and does not extend to other small CM weight-5 newforms.

## 11.3 Conjecture M13.1: Pollack-type rescue via Hodge-filtration framework

**Conjecture 2** (M13.1, ANT-tier, 2026-05-06)**.** *Let $f = f_{4.5.b.a}$ (LMFDB 4.5.b.a, $k = 5$, $N = 4$, CM by $\mathbb{Q}(i)$).*

*(a) A* Pollack-type rescue *for $f$ at $p = 2$ can be constructed via a Hodge filtration framework for critical-slope $p$-adic $L$-functions (cf. Kriz 2021,* Cam. J. Math *[bibkey TBD, not yet live-verified]): the anti-cyclotomic $p$-adic $L$-function $\mathcal{L}_2^{\pm}(f)$ on the anti-cyclotomic $\mathbb{Z}_2$-extension of $\mathbb{Q}(i)$ is well-defined at the critical slope, with the ± sign selecting the Heegner-point sub-tower.*

*(b) The functional-equation symmetrization ($\alpha_2 + \alpha_3 = 1/8$, M21) is a necessary condition for this construction to be self-consistent at $s = k/2 = 5/2$.*

*(c) The F1-renormalized sequence $v_2 = \{-3, -2, 0, +1\}$ (M22) interpolates the local Euler factor at $p = 2$ and provides the $p$-adic valuation datum for the anti-cyclotomic interpolation in Conjecture (a).*

The $N = p^2 = 4$ interlinking (§**??**) explains why all three parts of Conjecture **??** are required simultaneously: Findings 1, 2, and 3 of M13 are not independent; they form a **structural triplet** specific to the $N = p^2$ Steinberg-edge setting.

**Paper-2 status.** The working title is:

> *"The $N = p^2$ interlinking and a Pollack-type rescue conjecture for the CM newform 4.5.b.a"*

This is ANT-tier (targeting a math.NT journal; Conjecture M13.1 is well-posed, the statement is precise, and the three sub-parts each trace to live-verified LMFDB data). The F1 Euler-factor derivation (M22 recommendation) is the main remaining technical gap before submission.

**Joint DKSW–LVW programmes for $R$-2.1.** Two recent programmes advance the tools for Conjecture R-2.1. Dasgupta–Kakde–Silliman–Wang [**?**] prove Brumer–Stark over $\mathbb{Z}$, closing $p = 2$ ramified with integral precision via group-ring Hilbert Eisenstein and Ribet's method. Longo–Vigni–Wang [**?**] extend BDP Heegner cycles to algebraic Hecke characters of infinity type $(1 + \ell, -\ell)$, yielding an explicit Rubin formula outside the interpolation range. Neither covers ($K = \mathbb{Q}(i), p = 2$ ramified, $k = 5$ odd) directly; adapting DKSW to imaginary quadratic base fields and extending LVW to even discriminant and odd weight remains open (cf. M44.1(b), M75 F-1, M57). [TBD: prove]

## 12 Falsifiers: seven experimental disambiguators 2027–2045

The v7.5 amendment is falsifiable on seven near-future experimental timescales, listed in chronological order with target year, central-value prediction, and confidence-level threshold for refutation.

1. **NA62 $K \to \pi\nu\bar{\nu}$ + FLAG-2027 lattice $f_K/f_\pi$ on $|V_{us}|$ (2027–2028).** *[Downgraded from "v7.5 prediction" to "speculative observation" per A62+A72.]* The numerical coincidence $|V_{us}| = 9/40 = 0.22500$ (Section **??**) is retained as a **speculative numerical observation** ($P_{\text{null}} > 30\%$, A72; full-CKM null $< 1.7\sigma$ vs random, A62) but is **NOT** an ECI structural prediction. NA62 + FLAG-2027 tested at $\sim 0.3\%$ precision by 2028: if $|V_{us}|$ falls outside $0.225 \pm 0.001$ at $3\sigma$, the coincidence is ruled out, but this does not constitute a falsification of ECI's modular-flavour core (only a numerological accident would be removed). Survival margin $\sim 0.5$–$1.5\sigma$.

2. **KamLAND-Zen final $\langle m_{\beta\beta}\rangle$ + nEXO 2030+ (2027–2030).** v7.5 prediction $m_{\beta\beta} \sim$ few meV (CSD$(1 + \sqrt{6})$ at NO with $m_1 = 0$, Section **??**). Current KamLAND-Zen upper bound $\langle m_{\beta\beta}\rangle < 28$–$122$ meV [**?**]. A measured $\langle m_{\beta\beta}\rangle > 50$ meV at $3\sigma$ from KamLAND-Zen 2027 / nEXO 2030+ refutes the NO + $m_1 = 0$ component of A14 $\Rightarrow$ **REFUTED** (CSD$(1 + \sqrt{6})$ row).

3. **JUNO 2030 NMO at 3–4$\sigma$ (2030).** v7.5 prediction normal ordering with $m_1 = 0$ (Section **??**). Inverted ordering at $\geq 3\sigma$ by 2030 [**?**] $\Rightarrow$ **REFUTED** (CSD$(1 + \sqrt{6})$ row).

4. **DUNE 2030+ $\delta_{CP}$ (2030–2032).** v7.5 prediction $\delta_{CP} \approx -87°$ (Section **??**; CSD$(1 + \sqrt{6})$). DUNE 2030+ at $\pm 15°$ [**?**] on $\delta_{CP}$ rules out the prediction at $> 3\sigma$ if $\delta_{CP}$ centres at 0 or $+90° \Rightarrow$ **REFUTED** (CSD$(1 + \sqrt{6})$ row). Sensitivity sufficient by $\sim 2032$. The leptogenesis $(n-1)^2 = 6$ signature (Section **??**) inherits this falsifier indirectly through the joint $(\theta_{12,13,23}, \delta_{CP}, m_2, m_3, Y_B)$ global fit: if the seven-observable $\chi^2$/dof prefers $n = 1+\sqrt{6}$ over $n = 3$ at $> 2\sigma$ by 2030+, A14 + A55 are corroborated; if $\chi^2$/dof degrades by $> 3\sigma$ relative to $n = 3$, $\Rightarrow$ **REFUTED**.

5. **CMB-S4 $\Sigma m_\nu$ at $\sim 0.04$ eV ($\sim 2032$).** v7.5 prediction $\Sigma m_\nu \approx 0.06\,\text{eV}$ (CSD$(1 + \sqrt{6})$ NO floor, Section **??**). CMB-S4 [**?**] 2032+ at $\sigma(\Sigma m_\nu) \sim 0.04$ eV discriminates NO + $m_1 = 0$ (preferred prediction 0.058 eV) from inverted with $m_3 = 0$ (0.10 eV) at $\sim 2\sigma$. A measured $\Sigma m_\nu > 0.10$ eV at $3\sigma \Rightarrow$ **REFUTED** (CSD$(1 + \sqrt{6})$ row).

6. **LHCb-U2 / Belle II run 3 on $|V_{cb}|^2$ (2030+).** *[A62 RETIRED: $|V_{cb}|^2 = 1/600$ is no longer an ECI v7.5 prediction.]* The v7.4 entry $|V_{cb}|^2 = 1/600$ is RETIRED per A62: HFLAV-2025 average $|V_{cb}| = 0.04183(79)$ places 1/600 at $1.26\sigma$ (was $< 0.1\sigma$ on HFLAV-2024). The improved candidate $|V_{cb}|^2 = (7/4)\alpha_1^3 = 7/4000 = 0.00175$ at $0.0038\sigma$ (A62) is noted for record but *not* promoted to prediction (null test, A62+A72). This falsifier row is **VOID** for v7.5; it is kept for audit continuity with v7.4 only. Belle II 2027 at $\sim 0.5\%$ and LHCb-U2 / Belle II run 3 (2030+) at $\sim 0.2\%$ precision will constrain $|V_{cb}|^2$ but are not ECI discriminators at this stage.

7. **Hyper-K + DUNE proton-decay branching ratio (2030–2045).** v7.5 prediction (post-G1.12.B $M_6$) $B(p \to e^+\pi^0)/B(p \to K^+\bar{\nu}) \in [0.3, 3]$ at 95% CL (Section **??**). Outside $[0.3, 3]$ at $> 3\sigma$ by Hyper-K + DUNE combined 2030+ data, OR $\tau(p \to e^+\pi^0) > 3\times10^{35}$ yr at $> 2\sigma \Rightarrow$ **REFUTED**. Decisive Hyper-K 20-yr measurement $\sim$ 2045. This is the sole Nobel/Foundational-class testable prediction in the portfolio; G1.12.B M3–M6 closure (3.5–5 sub-agent months) is the immediate work-front.

To these we append a continuing v7.4 falsifier:

**(8) NMC cosmological NULL at Cassini.** A future improvement of the Cassini bound to $|\gamma - 1| \leq 10^{-6}$ that excludes the metric-NMC NULL falsifies the cosmological branch of v7.5-metric (but NOT v7.5-P; see Section **??**).

# 13 A72 honest negative result: Damerell-ladder numerology indistinguishable from random on fermion observables

This section records a *clean, definitive negative result* (A72, 2026-05-06) following the scientific norm of pre-emptive disclosure. Negative results of this type strengthen, rather than undermine, a research programme by eliminating unpromising directions and sharpening the surviving claims.

**Test setup.** The A72 null test constructed the full Damerell-derivable invariant space for $K = \mathbb{Q}(i)$: 150 candidates built from all products $\alpha_i \cdot \alpha_j$, $\alpha_m \cdot \pi^k$, $\alpha_m \cdot \sqrt{n}$ (for $n \leq 10$), $\alpha_m \cdot |L_p|$, and $w_K$ Weierstrass factors. These were compared against 19 low-energy fermion observables: PMNS angles $(\theta_{12}, \theta_{13}, \theta_{23})$, Dirac phase $\delta_{CP}$, Jarlskog $J_{\text{PMNS}}$, sum $\Sigma m_\nu$, ratio $\Delta m^2_{21}/\Delta m^2_{31}$, charged-lepton ratios $m_\mu/m_e$, $m_\tau/m_\mu$, CKM Jarlskog $J_{\text{CKM}}$, and quark Yukawa ratios $m_u/m_d$, $m_s/m_d$, $m_c/m_s$, $m_b/m_c$, $m_t/m_b$, $y_u/y_t$, $y_c/y_t$, $y_d/y_b$, $y_s/y_b$ (PDG 2024, NuFit-5.3, HFLAV 2025). For each invariant–observable pair, a rational quotient $q = \text{inv}/\text{obs}$ is computed; a hit is counted if $|\sigma_{\text{hit}}| < \text{threshold}$ and $|q| \leq q_{\max}$.

**Headline numbers.**

| Cut | Threshold | $Q(i)$ hits | Null mean ± sd | $\sigma$ vs mean |
|---|---|---|---|---|
| Lax | $\sigma < 0.5$, $\lvert q\rvert \leq 5/4$ | 718 | $698.5 \pm 42.9$ | +0.46 |
| **Strict** | $\sigma < 0.3$, $\lvert q\rvert \leq 1$ | **358** | $366.2 \pm 25.0$ | **−0.33** |

Null distribution: 512 Monte Carlo trials replacing the $K = \mathbb{Q}(i)$ Damerell ladder with random small-rational ladders of comparable magnitude (seed 20260505, deterministic). Strict-cut tail probability $P_{\text{null}} = 0.63$: $Q(i)$ is *below* random mean and wholly consistent with chance. Full posterior in `A72_DAMERELL_EXTENSION/null_test_results.json`.

**Verdict. STRONG DEAD-END.** The $K = \mathbb{Q}(i)$ Damerell-ladder numerology is statistically indistinguishable from random rational ladders on the full PMNS+lepton+quark observable space. This is a stronger result than A62 ($1.7\sigma$ on 2 CKM observables) because it covers 19 observables and 150 invariants with a 512-trial null distribution.

**What this kills and what it preserves.**

- **Killed:** any claim that the Damerell ladder $\{\alpha_m\}$ *predicts* or *structurally anchors* low-energy fermion observables ($|V_{us}| = 9/40$, $|V_{cb}|^2 = 1/600$, PMNS-sector $\alpha_m$ claims). The $H_7'$ axiom status is downgraded from structural input to notation convention (Section **??**).

- **Preserved:** the LMFDB 4.5.b.a CM newform anchor, Hecke $H_1$ restriction ($\chi_4$ nebentypus), and the Cardy hits $\alpha_2 = 1/12$, $\alpha_3 = 1/24$ as *mathematical* identities verified at dps = 60. These are arithmetic anchors, not predictions. The remaining six falsifiers (items 2–5, 7, 8 of Section **??**) are unaffected.

# 14 SUSY MSSM NO-GO decision (O1 D2, v7.6 new)

Sub-agent O1 (Opus 4.7, max-effort, 2026-05-06) delivered a definitive MSSM GO/NO-GO verdict:

**Verdict: NO-GO (Option B) with split-SUSY conditional.** ECI's modular sector at $\tau \approx i$ + A73 RG-stable wedge $\xi \in [-0.029, +0.001]$ + A55 high-scale see-saw thresholds operationally substitute for low-energy MSSM. Embracing the full MSSM would *triple* the parameter count and undermine the modular-minimality narrative.

**Quantitative arguments.**

1. ECI has $\sim 6$ free parameters in the lepton sector (NPP20) and $\sim 4$ in the quark sector (G1.12.B M1–M2 fitted); the MSSM soft-breaking sector adds $\sim 100+$ parameters.
2. The GUT-scale unification $\alpha_s$ prediction is adequate at 2-loop in minimal SU(5) without MSSM (O1 D1).
3. High-scale seesaw (A55 CSD$(1+\sqrt{6})$ at $M_R \sim 10^{14}$ GeV) provides the baryon asymmetry without low-energy superpartners.
4. The modular symmetry $\widetilde{\Gamma}(4)$ at GUT scale is generically broken by SUSY-breaking; introducing soft-SUSY terms would require a dedicated modular stabilisation analysis (not yet done).

**Publishable defence wording (O1 verbatim):**

> *"ECI adopts a no-MSSM stance on the grounds of theoretical economy: the modular $\widetilde{\Gamma}(4)$ symmetry at $\tau = i$ provides the fermion hierarchy without low-energy supersymmetry, and the A73 RG-stable wedge $\xi \in [-0.029, +0.001]$ shows stability without superpartner contributions. We do not claim the MSSM is ruled out experimentally; we claim it is superfluous within the ECI framework."*

**Split-SUSY conditional.** Migrate to split-SUSY (Arkani-Hamed–Dimopoulos 2004 framework) only if *both*:

- Hyper-K (2030+) confirms $\tau(p \to e^+\pi^0) \sim 6 \times 10^{34}$ yr (consistent with split-SUSY GUT threshold corrections); *and*
- $\alpha_s$ precision at 2-loop in minimal SU(5) fails to close by $> 2\sigma$.

Vanilla MSSM (low-scale, $m_{\rm SUSY} \sim 1$ TeV) is **excluded** from the ECI framework regardless.

# 15 Honest tensions audit (M2 falsifier matrix, v7.6 new)

## 15.1 Wolf 2025 $\xi$ tension

ECI $\xi_\chi^{\rm eff} = -0.024$ vs Wolf 2025 $\xi = +2.31$: a 5 orders-of-magnitude difference in the cosmological $\xi$ parameter. This is *non-falsifying* at the current data level: the two models use the same

cosmological dataset but differ in the underlying NMC Lagrangian and the prior on $\xi$. Resolution via Bayes contest: see §**??**. The ECI value passes the King–Gower gate; Wolf 2025 fails it in the metric formalism.

### 15.2 $\Sigma m_\nu$ tension

ECI v7.5–7.6 predicts $\Sigma m_\nu \approx 0.06$ eV (CSD$(1+\sqrt{6})$, $m_1 = 0$ normal ordering, Section **??**). DESI DR2 (2025, ΛCDM-only analysis) reports a 95% upper limit $\Sigma m_\nu < 64.2$ meV $\approx 0.064$ eV (in ΛCDM). Status:

- In a ΛCDM context, ECI's prediction is at the $\sim 1.7\sigma$ tension with the DESI DR2 UL (barely inside).
- In $w_0 w_a$CDM (which DESI DR2 prefers), the neutrino mass bound is substantially relaxed and ECI's 0.06 eV prediction is fully consistent.
- CMB-S4 (2032+) will resolve this at $4\sigma$ if the true $\Sigma m_\nu < 0.04$ eV (NO prediction killed); if $\Sigma m_\nu \in [0.05, 0.07]$ eV is confirmed, ECI is corroborated.

### 15.3 Alternative CM newform: 144.5.g.a

M19 identified 19 alternative CM weight-5 newforms as potential $H_6$ anchors. The leading alternative 144.5.g.a ($N = 144 = 12^2$, CM by $\mathbb{Q}(\sqrt{-3})$) passes Turing criteria T2 and T3 but **fails T1** by an analytic Euler factor argument: at $p = 2$, $a_2 = 0$ (the prime 2 ramifies in $\mathbb{Q}(\sqrt{-3})$ with Hecke character ramified at $\pi_2$, giving the $a_2 = 0$ case analogous to 11.3.b.a in M20's weight-3 check). A PSLQ period-ratio test is **PENDING** (HIGH-PRIORITY future action: see §**??**). Until the PSLQ test is run, 4.5.b.a remains the unique anchor by T1+T2+T3.

## 16 Outlook and open questions

We collect the open questions and conditional research directions that remain after v7.5.

**Q1: dMVP26 quark CP-phase compensator (Section ??).** Quantitatively, does a G1.12.B M3–M4 RGE matching reproducibly displace the modulus to $\mathrm{Re}\,\tau \sim 10^{-3}$ at the GUT scale, recovering the observed Jarlskog $J^q_{CP} \sim 3 \times 10^{-5}$ via the linear scaling of [**?**] p. 9? Alternatively, does the lepton-sector CP-spurion route (a) feed the CKM phase via a GUT-scale Yukawa-mixing term? This is a 12–16 week sub-agent campaign post-G1.12.B M3–M4 closure.

**Q2: BDP vs. second-form discrimination of $c \in \{7/10, 4/5\}$ (Section ??).** Either the BDP $p$-adic anti-cyclotomic complement supplies these unfit central charges via Iwasawa-algebra interpolation, or the v7.5 anchor must extend to a two-form anchor $(f_{4.5.\mathrm{b.a}}, f_{16.5.c.a})$. Discriminating (a) from (b) is the next concrete arithmetic test.

**Q3: A54 BC×CM scoping memo (Section ??).** 4-week Sonnet sub-agent to formalise the conditional math-ph note (publishable in 6–8 weeks if the chain is at least *conditionally* closed). Q1 2027 target.

**Q4: v7.5-P Palatini sub-branch decision matrix (Section ??).** The Palatini sub-branch makes four new predictions (P1–P4) under the assumption that the Wolf et al. 2025 large-$\xi$ signal [**?**] is real:

- P1: under a Palatini posterior, $\xi_\chi \in [+1.5, +3.0]$, $\log B > +5$ vs. ΛCDM.

- P2: $G_{\text{eff}}(z=0)/G_N \in [1.4, 1.8]$ in DESI Y3 + LSST 2030.
- P3: $|\gamma - 1| < 10^{-9}$ at 1 AU (Palatini Yukawa screening).
- P4: all v7.5 arithmetic-modular falsifiers (DUNE $\delta_{CP}$, Belle II $V_{cb}$, Hyper-K $p$-decay, CSD$(1+\sqrt{6})$, Damerell ladder) preserved unchanged.

Decision matrix:

- Wolf signal independently confirmed by 2027 (e.g. ACT DR6 + DESI Y3) $\Rightarrow$ COMMIT v7.5-P as primary cosmological branch.
- Wolf signal refuted (e.g. ACT DR6 finds no $4\sigma$ enhancement) $\Rightarrow$ DROP v7.5-P; v7.5-metric remains baseline.
- All v7.5 arithmetic predictions close NEGATIVE by 2030 $\Rightarrow$ DROP entire programme; v7.5-P moot.

**Q5: Leptogenesis sharper falsifier (Section ??).** The $(n-1)^2 = 6$ signature is indirect; a sharper falsifier requires either a derivation of the high-energy phase $\eta$ from ECI primitives (none known), or a cosmological probe of leptogenesis distinct from BBN $Y_B$. 4-6 week sub-agent campaign A56 in queue (NMC retrain Wolf-large-$\xi$ + leptogenesis joint).

**Q6: Modular-Higgs assignment as Axis-1 extension.** The Higgs doublet $H$ as an $S_4'$ non-trivial representation, in the sense of Penedo–Petcov 2020 [**?**] or Kobayashi–Tanimoto, would constrain $\lambda_H$ modular-form-wise. Cost $\sim$ 8–12 months sub-agent + Higgs-sector modular-form expert (Penedo, Petcov, King, Tanimoto). Foundational-Prize relevance LOW unless the prediction is at $< 5\%$ precision; current PDG $M_W/M_Z$ precision is 0.01%, so any modular prediction is unlikely to be falsifiable as new physics. We register this as a deferred Q3 2026 outreach item.

**Closing remark.** The v7.5 amendment consolidates the v7.4 architecture, retracts the LYD20 unified-scaffold graft per A46, and integrates four well-scoped grafts (Karam-Palatini, KW dS-trap, dMVP26 Kähler, A55 leptogenesis) plus two structural reframings (A49 two-tau, A54 BC$\times$CM honesty). A post-submission negative result (A72, Section **??**) establishes that the $K = \mathbb{Q}(i)$ Damerell-ladder numerology does not exceed random on the full PMNS+lepton+quark observable space ($\sigma = -0.33$, $P = 0.63$): the $H_7'$ axiom is downgraded to notation convention and the A17 CKM alignments are demoted to speculative observations. The surviving arithmetic core ($f_{4.5.\text{b.a}}$ LMFDB 4.5.b.a, Hecke $H_1$, $\chi_4$ nebentypus structure, Cardy identities $\alpha_2 = 1/12$, $\alpha_3 = 1/24$) is unaffected. The immediate work-front is G1.12.B M3–M6 closure (3.5–5 sub-agent months) toward the proton-decay branching ratio prediction at Hyper-K 2045+.

**Q7 (v7.6 new): 144.5.g.a PSLQ AFE computation.** The alternative CM newform 144.5.g.a passes T2+T3 but fails T1 analytically. HIGH-PRIORITY: run PSLQ period-ratio test at dps=60 against $\Omega_K^{(144.5.g.a)}$ vs $\Omega_K^{(4.5.b.a)}$ to definitively rule out 144.5.g.a as the $H_6$ anchor (§**??**). Estimated: 2-week Sonnet sub-agent.

**Q8 (v7.6 new): v7.7 production pipeline refinements.**

- Replace approximate $\theta_{\text{MC}}$ with exact CAMB-accurate computation (§**??**).
- Integrate full Planck 2018 off-diagonal covariance matrix into CMB likelihood.
- Run Wolf–KG production posterior ($\xi \in [+1, +4]$ prior) for 3-way Bayes contest (§**??**).
- Complete Adelic Katz paper-2 F1 Euler-factor derivation (§**??**).

**Q9 (v7.6 new): math.NT paper-2 submission path.** Conjecture **??** (Adelic Katz rescue, M13.1) is ANT-tier. The submission path is: (i) prove F1 Euler-factor derivation ($\sim 4$ weeks); (ii) write 10–15 pp math.NT note; (iii) post to arXiv math.NT; (iv) target journal: *Acta Arithmetica* or *Journal de Théorie des Nombres de Bordeaux*. Target: Q3 2026.

**Q10 (v7.6 new): Phase 3.F deep mathematical dives.** Following O1 D3 recommendation: Templeton OFI application (deadline 14 Aug 2026) to fund formal work on three Millennium Problems touching ECI structure:

- **Hodge conjecture:** ECI's modular structure suggests arithmetic Hodge classes in the motive of $f_{4.5.b.a}$; Phase 3.F scoping (Opus, 2 weeks) planned.
- **Riemann hypothesis:** GRH assumed throughout for $L(f_{4.5.b.a}, s)$; a formal GRH-conditional statement in paper-2.
- **Yang–Mills mass gap:** tangential to ECI; dropped from immediate scope per O1.

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KR thanks the Sonnet sub-agent runs A46 (LYD20 graft retraction), A47 (KW dS-trap compatibility), A48 (dMVP26 Kahler at $\tau = i$), A49 (Du–Wang two-tau autopsy), A50 (Karam-Palatini Cassini reference), A54 (BC×CM at $\beta = 2\pi$ scoping), A55 (CSD$(1 + \sqrt{6})$ leptogenesis), A36 (G1.12.B M5 proton-decay partial widths), A62 (CKM-Damerell systematic null test), A72 (full PMNS+lepton+quark Damerell-ladder null test, definitive dead-end, vectorised 0.8s on 4 CPU cores), A79 (post-CLPW H1 type-II$_\infty$ FRW literature review, 22 papers live-verified, 8-subclass partition), and S5 (H1 8-subclass partition graft into v7.5 §3, 2026-05-06). Phase 3.E sub-agents (v7.6 new): A71 (ECI NUTS production posterior, DESI+Pantheon+Planck), A73 (RG running of $\xi_\chi$, M20 (LMFDB Steinberg-edge cross-check, 8 forms), M21 (functional-equation symmetrization), M22 ($\beta$-renorm monotonicity + $N = p^2$ meta-finding), M23 (B-ratio independent-path audit, $B_{\text{Higgs}} = 1.077$), M24 ($\xi_\chi$ rail diagnostic, widened-prior confirmation), O1 (GUT 24-Higgs + SUSY NO-GO + Wave-12 meta-synthesis, Opus 4.7 max-effort), M25 (v7.6 synthesis, this document, Sonnet 4.6). Computational resources: PC RTX 5060 Ti (cosmopower-jax 386 predictions/s, JAX named_shape patch, 4-chain NUTS 1133 s). LMFDB queries: 4.5.b.a $L$-values and newform data cached 2026-05-05–06.

## Data and code availability

Repository: `https://github.com/AIdevsmartdata/crossed-cosmos`. Snapshot of v7.4–7.5 axioms: Zenodo DOI `10.5281/zenodo.20030684` (commit 6.0.53.7, 2026-05-05). v7.5-specific evidence: A46 `A46_LYD20_GRAFT/lyd20_pinned_results.json` (99,061× penalty); A47 `A47_KW_DSTRAP_COMPAT/` (KW `arXiv:2310.10369` live-fetch); A48 `A48_DMVP26_KAHLER_TAU_I/` (`kahler_tau_i.py`, `diag.py`); A50 `A50_KARAM_PALATINI/` (`section2_patch.tex`, `v75P_sketch.md`); A54 `A54_BC_CM_BETA_2PI/SUMMARY.md` (CMR05 `arXiv:math/0501424` chain analysis); A55 `A55_LEPTOGENESIS/` (`lepto_eta_B.py`, `v75_section_lepto` `lepto_summary.json`); A36 `A36_G112B_M5/` (`proton_decay_widths.py`, `tau_p_results.json`); A62 `A62_CKM_SYSTEMATIC/` (`null_test.py`, `ckm_hits.json`, `ckm_headline.json` — CKM-Damerell null $1.7\sigma$; $|V_{cb}|^2 = 1/600$ RETIRED); A72 `A72_DAMERELL_EXTENSION/` (`null_test_optimized.py`, `null_test_results.json` — full PMNS+Q+L null, $\sigma = -0.33$, $P = 0.63$, 512 trials, 0.8s). v7.4 baseline: cf. v7.4 amendment v2 [**?**].

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